

Chapter 9

The Impact of VLSI on CAM

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I. INTRODUCTION

In this chapter, as the title indicates, we shall discuss the impact that VLSI (very large scale integration) has and will have on manufacturing.

The whys and wherefores of the impact of VLSI on manufacturing can best be appreciated by a perfunctory discussion of manufacturing and of the entity referenced by the term "computer-aided manufacturing," or CAM.

Not long ago, the prominence of the United States as a leader in the ability to produce, not in just quantity, but also in quality was unquestioned. "You've come a long way baby," a phrase popularized by a commercial advertisement, is equally applicable to the advancement of technology in the United States. Stimulated and directed by sociological, economic, and political factors, our accomplishments helped develop the United States from a position of an emerging nation to one of the technological supremacy in less than 200 years. The motivation varied from those technological needs for transportation and industrial development of a young nation bent on territorial expansion to subsequent development of sophisticated communications system to bring our nation of people closer together. In the 1900s, the stimulus unfortunately resulted in most part from the needs of war, but this too promoted rapid technological development in many fields. Finally, political forces among others committed us to a race to outer space, leading also to continued achievements beyond anything humanity had yet experienced.

The two decades after World War II (1948–1968), as an example, saw average gains in rates of productivity of 3.2% and a steady rise in the standard of life. Stimulating this growth rate was the influx of innovations to manufacturing techniques and procedures. Innovations in technology alone are credited with supplying 40% of the average yearly productive growth increase during the two decades after World War II. The United States is credited with producing 60% of all worldwide technical innovations in the past 25 years. These decades and previous decades were a time of pride for Americans, pride as craftsmen and innovators in the products they produced and with which they supplied the world. Unfortunately, the story turns bleak starting in the late 1960s early 1970s and remains so to the present. Average yearly increases in productivity growth rates of only 2% were realized, with 0.5% in 1978 and 0.9% in 1979. The U. S. rates of productivity growth increases have not attained the growth rate realized during the two decades after World War II. The importance of continued growth is expressed in the 1978 *Economic Report of the President of the United States*, where the U. S. Council of Economic Advisors states that the slowdown in productivity growth in the United States is "one of the most significant economic problems of recent years."

This lagging growth rate of productivity is also a serious problem because of the fact that nations such as Japan and West Germany have

annual growth rates that surpass those of the United States. Figure 1 shows graphically the relationship between the U. S. and other major countries with respect to productivity growth.

Before continuing, an important point needs to be made; even though during 1979 the rate of productivity growth increase in the United States decreased and that of countries like Japan and West Germany increased, the United States is still the most productive country in the world; but in order to remain the most productive country, we must mend the ills of our fossilizing manufacturing centers.

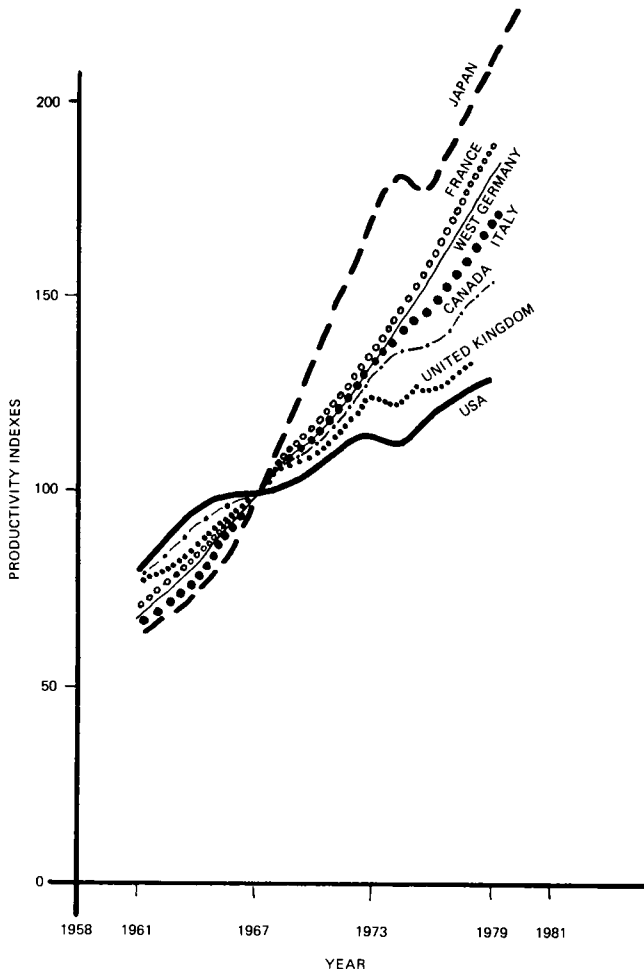


Fig. 1. Productivity growth (U. S. Department of Labor, Bureau of Labor Statistics, compiled by Dr. C. W. LeMaistre).

Manufacturing structures that were developed during an era of cheap and plentiful sources of energy and natural resources were not renovated to reflect a changing world situation, resulting in inherent inefficiencies in manufacturing with regard to energy and natural resource usage. An example of inefficiency would be the nonoptimization of the flow of material through a manufacturing process. Rather than routing a part on a pallet via a conveyor system, which moves the part quickly and smoothly from one process site to another, numerous setups are required to machine a part as it is moved from one location in a plant to another.

Today, fewer people want to work in manufacturing. In 1974, the United States had about 30% of all employed people working in manufacturing. The Bureau of Labor statistics estimate that by year's end the percentage of Americans working in manufacturing will have dropped to near 22% (see Fig. 2). The problem with which the United States is faced is a continual shrinking of the manufacturing work force. As stated in a paper by Mr. James A. Geier, President of Cincinnati Milacron, Inc. [1], by the end of this century, the Rand Corporation forecasts that perhaps 2% of the U. S. work force will be engaged in manufacturing. Talented individuals need to be streamed into manufacturing in order that the advanced technologies existing today, which need to be developed, may be inte-

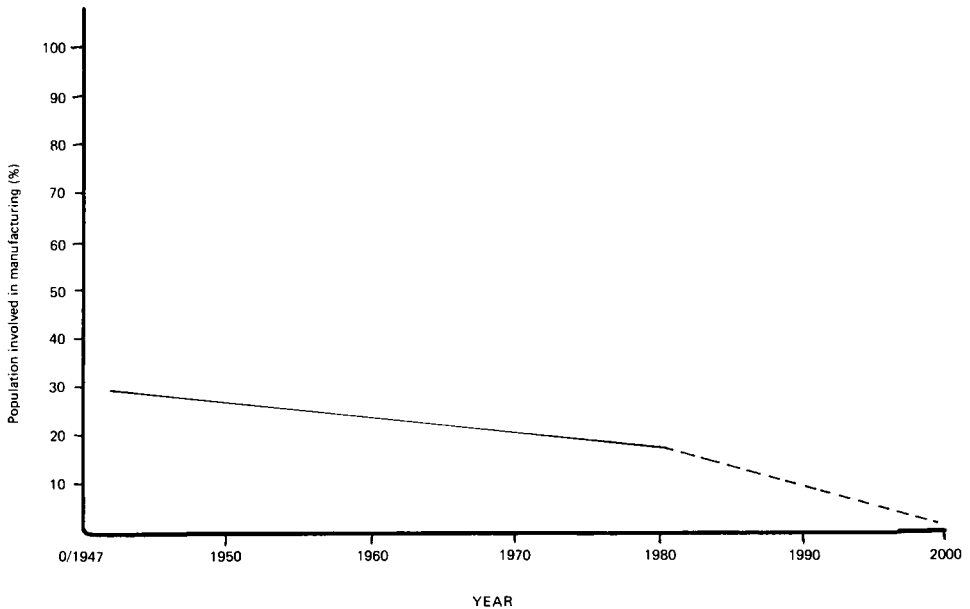


Fig. 2. Percentage of Americans working in manufacturing.

grated into manufacturing, the result leading to more efficient and productive manufacturing systems. The inimical attitude toward manufacturing needs to be amended. Also, the lack of a work force in manufacturing must be compensated for more by the integration of computers controlling manufacturing processes. Many manual tasks to be replaced by computer controls are tedious, dangerous, or of a hostile nature to the human worker; hence, even now it is hard to find a human to do many tasks, which necessitates that these tasks be carried out under computer control.

Countries such as Japan and West Germany have been excellent students of the United States manufacturing system. To this point, they have become the major suppliers in a worldwide market and in some instances more productive, as in the case of Japan and the automotive industry. The United States automotive worker produces about 25 cars/yr on the average compared to his Japanese counterpart who produces close to 49 cars/yr at about one-half the rate of pay with some ten times the amount of robots as are used in the United States. Japan is also threatening to dominate the semiconductor industry, an industry that we have developed almost exclusively and have dominated since its infancy.

There are indications that the United States may be losing its competitive edge on an international basis even in this industry. In 1979, for example, the semiconductor business represented over \$10 billion of worldwide sales, double the figure for 1974. Another doubling is forecast for 1984 when sales are estimated to exceed \$21 billion. However, in 1973, we exported \$120 million semiconductors to Japan and imported only \$20 million, a 6:1 ratio; whereas in 1979, we exported some \$380 million semiconductors to Japan compared with \$270 million imported, a 1:4:1 ratio. More specifically, with computer memories as the largest-selling integrated circuit, the Japanese companies, Nippon Electric, Hitachi, and Fujitsu, already represent three of the top seven suppliers and sell 42% of the world supply with the other four U.S.-based companies being Mostek, TI, Motorola, and Intel. The changes, or trends, in such figures are even more dramatic when citing the shift of technological supremacy. The semiconductor industry has been one of our major successes, in which we have always led the field internationally, based on creativity, innovation, aggressiveness, and productivity. The picture is changing.

Other countries have been successful because of social factors (e.g., homogeneous population), governmental structures and incentives (e.g., antitrust laws that are not as detrimental to industry as in the United States), and sheer applications of existing technologies to develop innovations in manufacturing (e.g., 45,000 robots used in manufacturing in Japan as opposed to 5000 in the United States).

Movement in the application of developed technologies and the development of new technologies is needed in manufacturing in an effort to develop more efficient, competitive, and productive manufacturing techniques, processes, and plant architectures. Such movements have been transpiring in manufacturing with a resultant increase in manufacturing efficiency and productivity. An example of an entity that has impacted manufacturing tremendously, in a positive manner, is the development and subsequent introduction of numerically controlled metal-cutting machine tools during the early 1950s. The greatest and most recent impact has been realized by the introduction of the digital computer to manufacturing. The impact of the digital computer is making manufacturing more efficient, and productivity has, as yet, not been fully realized. This influx of computers, and particularly the effect of VLSI on price and size of computers, into the manufacturing arena is the topic of this chapter; in particular, computer-aided manufacturing, or, more succinctly, CAM.

Just as there was an industrial revolution during the eighteenth century, resulting in the mechanization of manual labor, there is transpiring today another industrial revolution. This new revolution, characterized as a knowledge and information revolution, is being kindled by the revolution occurring in electronics.

II. CAM'S STRUCTURE

In this section we shall delve into what is meant by the acronym "CAM" and into its structure. This understanding is essential to an appreciation of what VLSI is and how it can have an impact on CAM.

An appropriate and fundamental starting point will be to give a working definition of CAM.

A computer-aided manufacturing (CAM) system entails a hierarchical structure of computers linked together to form an integrated information network that will monitor a broad spectrum of interrelated tasks and controls, each of these tasks based on some overall management strategy, with the flow of information being both upstream and downstream.

Figure 3 depicts the hierarchical CAM structure. As can be seen, the levels in our CAM structure are separated into four major areas: management, finance, engineering, and productivity, and under each of these major levels are various sublevels. The major levels and sublevels of our hierarchical structure represent independent computer systems that are linked together to form a network.

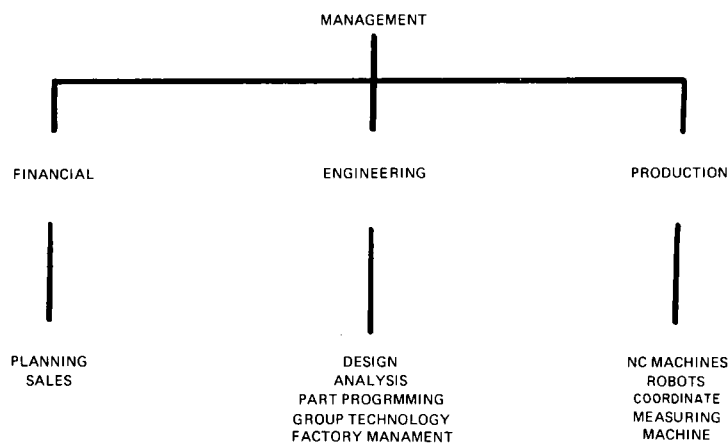


Fig. 3. Hierarchical CAM structure.

A. Management

In a top-down analysis of this hierarchy, we first consider the management level. This computer is the main data base storage, processing, and overall decision-making element of our structure. With respect to data bases, all archival and working information is stored here, and upon request from other levels, data can be transmitted to them as required. Included in this data base is vital information transmitted upstream from the various lower levels. This information is management information and will be processed by the management system. Based on the results of the information received and processed, the system will make management decisions concerning the operations under its control. Controlling instructions will then be transmitted to the appropriate level(s), depending upon information received, and the management strategy will be programmed into the highest echelon processor. The various management levels of the corporation can access the dynamic data bases and establish the status of

TABLE I
CAM DATA BASE

Engineering design data	Quality control data
Master production planning schedule	Machinability data
Management reports	NC production data
Process monitoring data	Process control data
Financial data	Inventory control data
Sales data	Maintenance data

the functional operations within their organization. A few of the structural elements that comprise a CAM data base are illustrated in Table I.

B. Finance

Finance is a major division within the CAM structure and entails such sublevels as accounting, marketing, financial planning with the aid of the financial modeling, and sales planning and forecasting, to name a few. This level, as can plainly be seen by the example depicted, deals directly with the flow of revenues of an organization and supplies information for the planning of business based on the financial posture of the organization and the needs of the marketplace.

C. Engineering

The next major level we come to is engineering. This level involves such entities as product design, drafting, and analysis, which fall into the realm of computer-aided design (CAD); research and development; product testing; manufacturing engineering, which entails such things as numerical control (NC) part programming and process planning. On this level, we are concerned directly with creation of the product planned by the previous two major levels of the CAM structure. This level is the precursor of the actual fabrication, or production, phase in our CAM environment, which we shall discuss next.

D. Production

Finally, but not the least important, is the production level. Production entails, not only the actual manufacturing of the product, but also the control of the manufacturing process. Included are such entities as inventory; distribution of raw materials (tools, etc.); production control; maintenance of manufacturing-related equipment and machine tools; robots; quality assurance; final assembly; and shipping.

At this point, we have completed our brief introductory remarks on the hierarchical CAM structure. These are the basic elements that comprise the CAM hierarchical network. It is through the network of processors, the transfer of information, the processing of this information, and the resultant management decision control outputs that the power of a CAM system resides.

III. VLSI'S IMPACT ON CAM

A. Management-Finance

Management at all levels requires information on the current status of affairs directly under their control in order to make useful, valid decisions. The more current the status of the required information and the more flexible a means by which individual managers may compare various parameters, the more efficient and effective manager and organization will result. Information system data bases have been developed in many organizations and have become tremendous assets to their users, but these systems are limited in their usefulness by the fact that comparisons of various data bases are fixed; the individual users cannot readily access and compare various data bases since, at present, this can require much software programming.

Beginning in the early 1960s, the availability of large computer systems allowed management to develop internal information systems for the purpose of accounting and financial control. Planning and controlling of production systems was an application that would come later. The initial information systems applications involved taming the already well developed heuristic accounting systems and placing them on the computer. The concept of a global data base or corporatewide data base containing all the information needed for managing the corporation quickly emerged and remained a dream for most of the 1960s and into the early 1970s. Corporations that attempted to build such data bases found that the ability to acquire correct and accurate data from a variety of sources in many different forms and retrieve the data in a way that would make it useful to managers at different levels with different functions was not economical. In the early 1970s, the availability of minicomputers changed the concept of the global data base to a decentralized series of data bases controlled by the functional departments to which they belonged. That is, marketing would have its own data base; engineering would have its own data base; and so on. As the functional organization developed its own data base for its own applications, corporate management quickly found that these data bases tended to be accurate and current because they were being compiled by the users for the users' specific needs. The corporatewide data base therefore became a series of decentralized, functional, organization data bases that were linked back to the mainframe of the corporate computer.

The impact of VLSI on this system is obvious; the limitation of the

individual data bases has always been the size and speed of the minicomputers on which they have been developed. The economy of using individual large computers on a decentralized basis was not available. Now, VLSI will allow small, relatively inexpensive, but very powerful, computers to be used by these functional organizations. This will allow the data base management concepts to be enhanced to include broader applications within the functional organization and, therefore, to build the effectiveness of the overall, or global, corporate data base. The availability of user-oriented software will allow an increasing number of people to develop and use the computer system. The access of the CRT ports and other more advanced peripheral devices to this system will allow a more accurate development of the data needed, both for the operation of the functional department and for the needs of the corporate as a whole. The functional managers' attention will be drawn more toward the exogeneous variables needed by the organization since they will have total control of the internal data.

An entity that already has had a tremendous impact on the business community is interactive computer graphics. Most succinctly, interactive computer graphics's impact is due to the fact that one picture is worth a thousand words.

The ability of the user to compare various parameters and to view the result in graphical form, rather than as reams of numbers, makes analysis of the comparison straightforward. Marked or subtle changes in the slope of a graph or the heights of a bar chart are quickly and easily noticed, whereas printing all the numbers required to create the charts in a matrix and then interpreting the results is tedious, if not impossible. Recent advances in interactive color computer graphics is yielding graphical results with new dimension. Color graphs or bar charts allow more detail to be more readily included and analyzed per chart.

Developments in VLSI will make available more local intelligence to individual user so that they may easily create, via conversation with the computer, a program that will compare any of the data bases residing on the system in the manner chosen by the user. This will yield, as already noted, more control of an organization to the managers by enabling them to have the required management information readily, in the most appropriate and, most importantly, current form.

B. Engineering

Engineering comprises product design, drafting, and analysis, finite-element analysis, numerical control part programming, and process plan-

ning. This level is the precursor of the actual fabrication, or manufacturing, phase in our CAM environment.

1. Computer-Aided Design

Another chapter deals with computer-aided design (CAD) in general. Here we shall briefly address the topic as it applies to engineering for manufacturing. It is an area that has been greatly impacted and changed because of VLSI, and one that has subsequently revolutionized CAM.

Prior to the advent of CAD, specifications for a part were supplied to a drafting group who would then create, by hand, the engineering drawing. This drawing would then be used by many groups of people in the organization, such as management, sales, and manufacturing engineering. Also,



Fig. 4. Interactive computer graphics for design (courtesy of Applicon).

as already noted in Section III.A, CAD has direct applications in the management–finance groups.

The application of CAD to engineering created a very powerful tool to be used in the area of design, drafting, finite-element analysis, and NC part programming. Engineering drawings, which, prior to CAD, required much labor to reflect design modifications (i.e., redrawing of the complete engineering drawing or attaching to the engineering drawing an appendage that referenced the change(s)) can now be done rapidly. The designer/drafter need only sit in front of an interactive computer graphics CRT and request that the engineering drawing that requires modification be displayed on the CRT (see Fig. 4). The operator can then interactively and rapidly make the required modifications to the engineering drawing and all users of the engineering drawing will immediately have an updated version available to them. A number of CAD/CAM turn key systems are being marketed today, and Fig. 5 depicts such a system marketed by Computervision, Inc.

The actual creation of the engineering drawing by use of a CAD system increases production. Figure 6 shows a finished part, which was designed



Fig. 5. CAD/CAM turnkey system (courtesy of Computervision, Inc.).

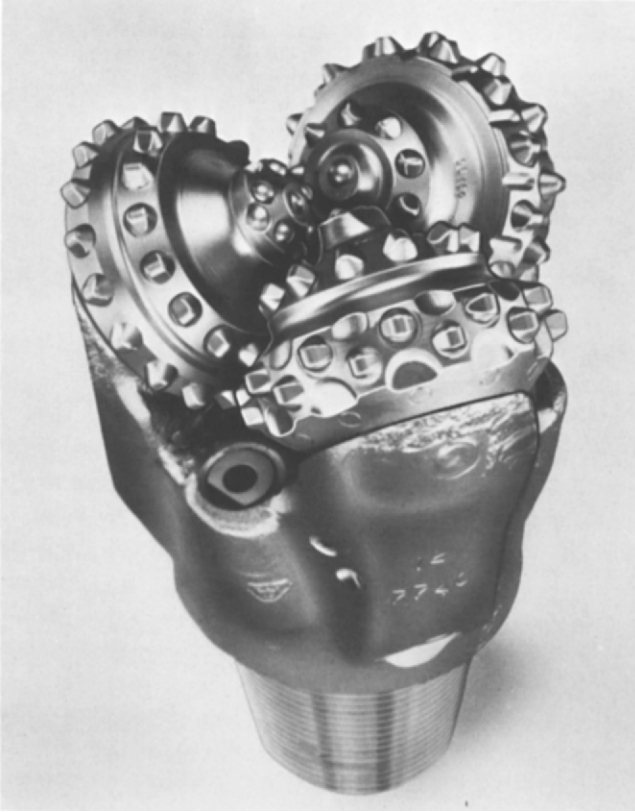


Fig. 6. Finished part (courtesy of Applicon).

by the user shown in Figure 7. The device used here is an Applicon Interactive Computer Graphics System.

Advances in many areas of technology are currently being impeded because effective and practical three-dimensional object representation schemes and associated algorithms for manipulation, analysis, and display are not available. Generally used techniques have a limited range of applicability because of shortcomings in two major areas. First, representation capabilities are not sufficiently robust to easily handle the object complexities required in a realistic environment. Second, manipulation and display algorithms, such as interference detection and hidden surface removal, require extremely large numbers of calculations to be applied in practical situations. They often exhibit exponential (typically quadratic) growth in the number and complexity of the objects. It is not believed that near-term progress in hardware technology will render such



Fig. 7. Designer using CAD/CAM system to create part in Fig. 6 (courtesy of Applicon).

schemes practical for the vast majority of potential applications. Much more efficient data structure and algorithms, which take advantage of current hardware technology, are needed.

Existing three-dimensional representation schemes have been divided into the following six categories by Voelcker and Roquicha [2]:

- (1) **Primitive instancing.** Families of objects are defined parametrically. A shape type and a limited set of parameter values specify a specific object.
- (2) **Spatial enumeration.** An object is represented by a list of the cubical spatial cells that it occupies.
- (3) **Cell decomposition.** A generalized form of spatial enumeration in which the disjoint cells are not necessarily cubical or even identical.
- (4) **Constructive solid geometry (CSG).** Objects are represented as collections of primitive solids (cuboids, cylinders, etc.). They are connected via the Boolean operation.

(5) Sweep representation. A solid is defined as the volume swept by a two- or three-dimensional shape as it is translated along a curve.

(6) Boundary representation. Objects are represented by their enclosing surfaces.

Specific advantages and disadvantages of each are tabulated by Voelcker and Roquicha [2], along with a classification of 21 existing systems. Most use CSG (TIPS, PADL, SYNTHAVISION, etc.) or boundary representation (BUILD, CADD, EUCLID, ROMULUS, etc.). In a separate study, Baer, Eastman, and Henrin [3] compiled an analysis and comparison of 11 popular systems.

The currently used systems are characterized by a limited domain over which objects can be represented. Mathematically well defined surfaces or solid primitives such as cuboids or cylinders are typically fitted to the desired shape. Some systems allow quadratic surfaces and higher-order patches to be employed, but the more general and more powerful primitives usually require substantial additional computations for object manipulation and display. Also, adding a new primitive to a system or generalizing the use of an existing one may necessitate extensive development of mathematical tools and significant software modification. Another consideration is the potentially large labor cost involved in the fitting of large numbers of primitives to a desired object. Three dimensions of two objects would require a representation of several thousand.

Object manipulation and display can easily place an unacceptable burden on computational resources. For example, calculating the edge of intersection in three dimensions of two objects represented by several thousand primitive shapes and surface patches can be a very formidable task. Hidden surface display and interference detection present similar problems. Exponential, rather than linear, growth can be expected.

At the root of the computational task is a sorting task. The primitive must, in effect, be paired off and then each pair analyzed, an experimental growth operation.

Some operations are linear in the number or complexity of the object but require a search of the object space many times. Visibility determination via ray firing, for example, can require a search at each pixel in the image (typically about two hundred fifty thousand or 1 million).

Some recent work has involved the segmentation of the universe or the enclosing of objects in boundary boxes so that, in performing an operation, some objects can be easily eliminated from consideration.

The implementation of fast and efficient application algorithms is often made very difficult by the structure of the data representation. A simple request for the approximate volume of an object or the approximate

location of its center of mass could require a very large number of calculations involving thousands of primitive objects.

Finally, the problem of implementation in hardware is often aggravated by the data structures and algorithms. Many techniques were designed when an evaluation of hardware situation called for compact data structures and algorithms that would fit into a limited memory (typically 64K bytes). Calculations were to be handled by a single serial processor, most often a general-purpose minicomputer. This strategy has resulted in schemes that are very efficient in memory usage but require many floating-point calculations.

A three-dimensional, geometric, modeling scheme and associated linear growth algorithms have been developed [4,5] so that objects of arbitrary complexity can be encoded, manipulated, analyzed, and displayed interactively in real time or close to real time in parallel, low-cost hardware. (Fig. 8) Specifically, the goals are as follows: (1) to develop a capability to represent any three- (or N -) dimensional object to any specified resolution in a common encoding format; (2) to operate on any object or set of objects with the Boolean operations (union, intersection, and difference) and geometric operations (translation, scaling, and rotation); (3) to implement a computationally efficient (linear) solution to the N -dimensional interference problem; (4) to develop the capability to display in linear time any number of objects from any viewpoint with color, shading, shadowing, multiple illumination sources, transparent objects, orthographic or perspective view, and smooth edges (antialiasing); and (5) to develop a

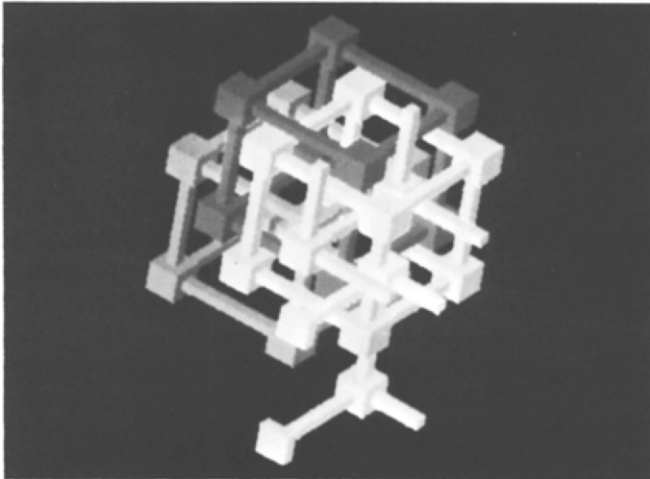


Fig. 8. Octree solid model.

scheme that can be implemented across large numbers of inexpensive, high-bandwidth processors, which do not require floating-point integer multiplication or integer division capability.

These goals have generally been met on the octree encoding scheme, as in Fig. 8. In addition to the preceding points, two additional favorable characteristics have been noted. First, the performance of the system degrades gracefully as the complexity of the situation increases. Second, the user has the ability to trade off computation and processing precision. This means, for example, that a coarse image can be generated very quickly with much higher fidelity details emerging over a period of time.

None of the algorithms developed to date (union, intersection, subtraction, translation, scaling, rotation, or numerous display functions) require floating-point operations or integer multiplications or division except the most general form of perspective display, which requires integer division. Thus, the algorithms can be implemented in relatively inexpensive, high-bandwidth hardware.

The main disadvantage of the encoding technique is the large memory requirements. Hunter and Steiglitz [6] present a proof that the memory required to store (and the calculations to process) a quadtree two-dimensional object is on the order of the perimeter of the object. It can be shown that the memory and processing computations required for a three-dimensional object are on the order of the surface area of the object. Depending on the object and the resolution, this can still represent a large memory-storage requirement.

In a particular situation, if ideal primitive objects (cubes, spheres, cylinders, etc.) are sufficient to accomplish the task, octree encoding is probably not the technique to employ because of the volume of memory required. If, however, the objects are to be deformed or have complex surfaces, the octree techniques could hold important advantages.

Another area where CAD has and will continue to have a tremendous impact is in the realm of VLSI design. Work in the area of VLSI design automation, layout optimization, and automatic test set generation is being done by several investigators and will be the basis for the next section.

2. Impact on VLSI Design by CAD

The design of optimized mask layout and the automatic generation of test patterns for very large scale integrated circuits are problems of acknowledged computational difficulty. The origins of this difficulty can be traced to the fact that both of these problems contain numerous sub-tasks. As a practical matter, this means that for certain (but not all) kinds

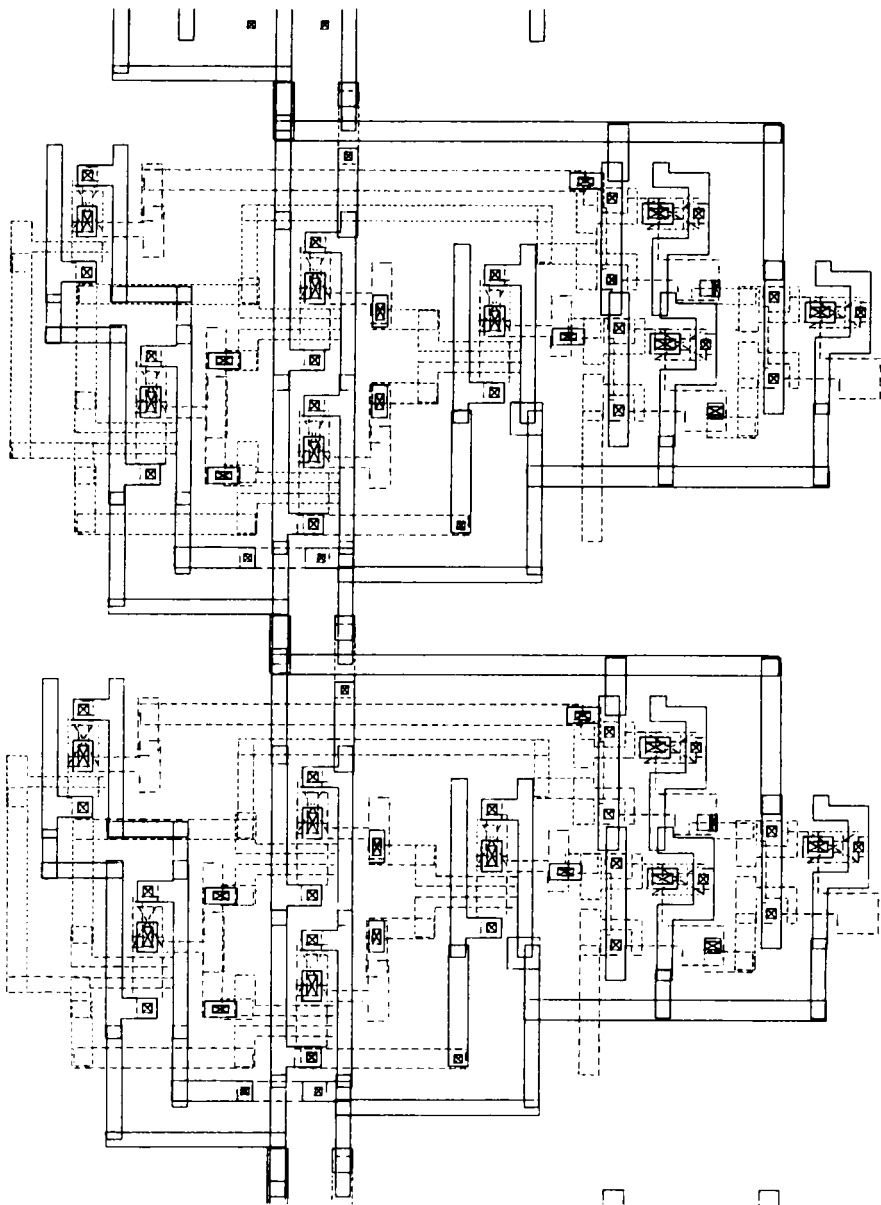


Fig. 9. Courtesy of Dr. J. F. McDonald of Rensselaer Polytechnic Institute.

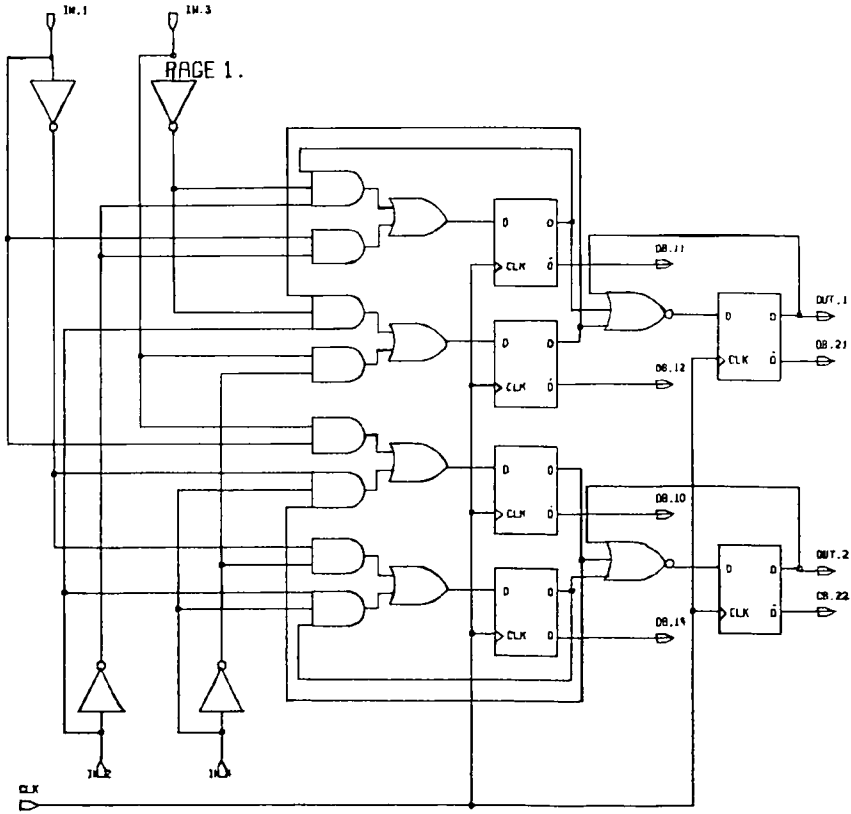


Fig. 10. Courtesy of Dr. J. F. McDonald of Rensselaer Polytechnic Institute.

of large designs, truly optimal layouts or completely comprehensive test sets may be very difficult or essentially impossible to obtain in a cost-effective manner. It is the goal of design automation and test set generation research to find for a layout or test set an approximately optimal solution that is achievable with a reasonable amount of computation. Many rigid optimization problems share the property that they lie extremely close to very workable schemes that can be employed to obtain these nearly optimal solutions for all the applications of interest. Finding such algorithms, however, is likely to require a gifted mix of practical programming methods and theoretical insight. What is needed is the development of a set of usable design and test set generation software tools that are interactive and provide for the inclusion of a variety of automatic procedures designed to speed up certain routine computations and to execute certain large global optimizations iteratively. See Figs. 9 and 10.

Design automation (DA) is a generic term covering all the activities associated with the creation of a digital system, including its architectural conception, implementation in integrated circuit mask layout, manufacturing, and, ultimately, testing. In theory, nearly all of the design steps associated with these activities could eventually be completely automated except for the conception of the design. In point of fact, as these designs grow in complexity, full automation of the various design steps will become mandatory. As these DA packages evolve, it is important to note that excellent compatibility must exist among the design, simulation, manufacturing, documentation, and testing phases to assure the usability and acceptability of the final system by its customers in the design community. This compatibility does not happen automatically, and a constant effort must be exerted to maintain compatibility of data bases. Alterations of design at any given step of the DA process must be reflected throughout all of the data structures in order to permit the back-referencing required in iterative design.

As an example, work on design automation has been conducted at Rensselaer (sponsored by General Electric and the U. S. Naval Air Engineering Center in 1978) in which initial efforts were concentrated on the development of a logic-schematic drafting package or logic entry system (LES). Similar in structure to the Stanford University drafting system (SUDS), the Rensselaer LES package was distinguished from its Stanford predecessor by being cast as a highly transportable FORTRAN subroutine, which could easily be called by any DA system component, thereby permitting back-referencing from any phase of the design process to the original logic schematic. In this way, any simulation, testing, or artwork generation could be coordinated with this schematic.

The purpose of the LES package was initially to assist in creating a benchmark data base, which could serve as a test bed for comparison of various automatic test pattern generation (ATPG) algorithms. This led quickly to the creation of a translator from the LES data base to an input format compatible with the popular, commercially available LASAR ATPG program, as well as a locally researched ATPG program, called the combinational circuit control algorithm. Later, the simple concept of a compatible format, back-reference reversible data base translator was extended to produce simulation and wirewrap prototype layout packages.

Current activity is concentrated on developing an automated, hierarchical macrocell-oriented NMOS mask layout program. The structure of this program, as currently envisioned, would produce mask artwork automatically or semiautomatically from four potential sources:

- (1) the original logic schematic,
- (2) "stick" MOS transistor diagrams,

- (3) module/layout drawings, and
- (4) hand-drawn artwork.

Currently, the capacity of the system is 2000 gates/design.

Partially in anticipation of growing demand and partially for educational objectives, an effort was begun early in 1980 at Rensselaer to create a convenient graphics-based program for facilitating artwork generation for custom and semicustom integrated circuit design. As a result of the popularization of NMOS design, resulting from the book by Mead and Conway [7], the first artwork effort undertaken was automatic production of an NMOS layout. Educationally, the standard artwork format, known as the

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GII TRANSLATION FOR CIRCUIT MULTI
      5 NO. OF INPUTS
      8 NO. OF OUTPUTS
      2 NO. OF LEVELS
      4 NO. OF FLIP FLOPS IN LEVEL 1
      2 NO. OF FLIP FLOPS IN LEVEL 2
*      FF LEVEL 1 GATE NO. 1 PIN NO. 1
      P1 Q11 P4 P2
*      FLIP FLOP STATEMENT
F111 P' 1Q11P'4 + P2P'4
*      FF LEVEL 1 GATE NO. 1 PIN NO. 2
      P5
*      FLIP FLOP STATEMENT
F112 P5
*      FF LEVEL 1 GATE NO. 2 PIN NO. 1
      P1 Q12 P4 P3
*      FLIP FLOP STATEMENT
F121 P' 1Q12P4 + P1P'3
*      FF LEVEL 1 GATE NO. 2 PIN NO. 2
      P5
*      FLIP FLOP STATEMENT
F122 P5
*      FF LEVEL 1 GATE NO. 3 PIN NO. 1
      P1 P2 P3 Z13
*      FLIP FLOP STATEMENT
F131 P1P2 + P3P'2Q13
*      FF LEVEL 1 GATE NO. 3 PIN NO. 2
      P5
*      FLIP FLOP STATEMENT
F132 P5
*      FF LEVEL 1 GATE NO. 4 PIN NO. 1
      P2 P3 P4 Q14
*      FLIP FLOP STATEMENT
F141 P'2P'3 + P3P4Q14
*      FF LEVEL 1 GATE NO. 4 PIN NO. 2
      P5
*      FLIP FLOP STATEMENT
F143 P5
*      FF LEVEL 2 GATE NO. 1 PIN NO. 1
      Q11Q21 Q12
*      FLIP FLOP STATEMENT
F211 Q'11Q'21Q'12
*      FF LEVEL 2 GATE NO. 1 PIN NO. 2
      P5
*      FLIP FLOP STATEMENT
F212 P5
*      FF LEVEL 2 GATE NO. 2 PIN NO. 1
      Q13 Q22 Q14
*      FLIP FLOP STATEMENT
F221 Q' 'Q'22Q'14
*      FF LEVEL 2 GATE NO. 2 PIN NO. 2
      P5
*      FLIP FLOP STATEMENT
F222 1'5
*      OUTPUT 1
      Q14
      ...

```

Fig. 11. Intermediate symbolic output for artwork generator.

California intermediate form (CIF), has made possible the production of student-designed integrated circuits that could be submitted under the "multiuniversity" wafer program for modest costs. Hence, a goal of the project was production of CIF output. The translator for producing this from artwork has already been completed. Hence, once the artwork production program NAS-I is fully completed, Rensselaer students can join the "multiuniversity" wafer program using some of the most powerful graphics design aids available to any school in that program. Translators-to-artwork formats other than CIF (such as the Applicon format) are achievable. Fig. 11 illustrates an intermediate symbolic output to be passed on to the artwork generator.

PLA translation will help relieve metallization congestion, but remaining metal and polysilicon interconnection conflicts will still require the use of module placement and channel routing algorithms. New channel routing algorithms providing for over-the-cell intraconnect paths by Deutsch are being considered for this phase. Algorithms for placement are in the public domain at Lawrence Livermore Lab. These are the only nonproprietary placement packages available at low cost to universities in the form of actual source code. Obtaining a source code for these programs is obviously helpful in creating new systems. However, transportability for such software is low. Some new ideas based on clustering algorithms of image processing are being studied. Integration of some of the newer placement routines (such as COMET) [8] or routers (such as MP2D) would also be extremely helpful.

C. Manufacturing Engineering

Two major duties encompassed by the manufacturing engineering group are numerical control (NC) tool-path generation (or NC part programming), which is a program using an engineering drawing as a basis to guide the motion of a machine tool as a part is being cut, and the production of a process plan by which the flow of material through the factory floor is controlled, both of which are impacted significantly by VLSI.

1. NC Part Programming

An examination of Fig. 12 depicts the trends through the years by which innovations in electronic technology have changed the methods by which NC part programmers create an NC tool-path program.

The introduction of NC machine tools into the manufacturing environment around 1957 saw machine tools whose part programs (note: a "part program" is synonymous with the term "tool-path program") were

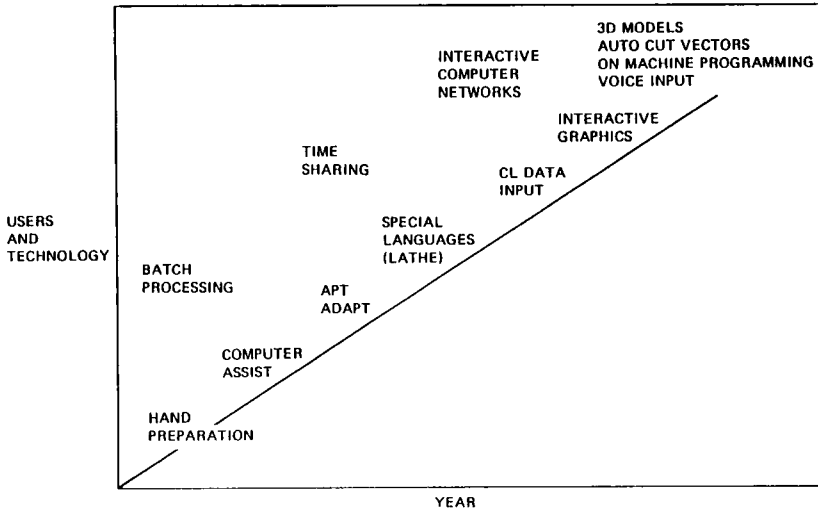


Fig. 12. Trends in control input methods.

written by the programmer in an assembler-type format without the aid of a computer. The part programmer would, from the engineering drawing of the part, give the coordinator of the point for the tool in its desired position. The medium used to transfer this information to the machine tool was punched tape.

There exist two types of NC machine tools, a point-to-point machine tool and a continuous-path machine tool. On a point-to-point machine tool, the part programmer need only define the coordinates at which a function is to be performed (e.g., drilling) and need not be concerned with the path that the tool takes in arriving at that location, since between intermediate points no metal is cut. The only concern the programmer has is that between intermediate points the part, clamps, or fixtures are not violated. It is the machine controller that will move the tool and, hence, will not violate the machine itself. The second type, the continuous-path machine tool, is of greater complexity to program. On this type (e.g., mill), the tool is in continuous contact with the part as the tool moves from one coordinate to the next.

Reiterating, the part programmer in the early days wrote the part program in an assembler-type format without the aid of the processing powers of a digital computer. To create the part program, the programmer viewed the engineering drawing and juxtaposed a coordinate system onto the part to represent the part's location on the machine-tool table (see Fig. 13). Then the programmer proceeded to label the coordinates of strategic points on the part. If a point-to-point machine were to be used, say, for

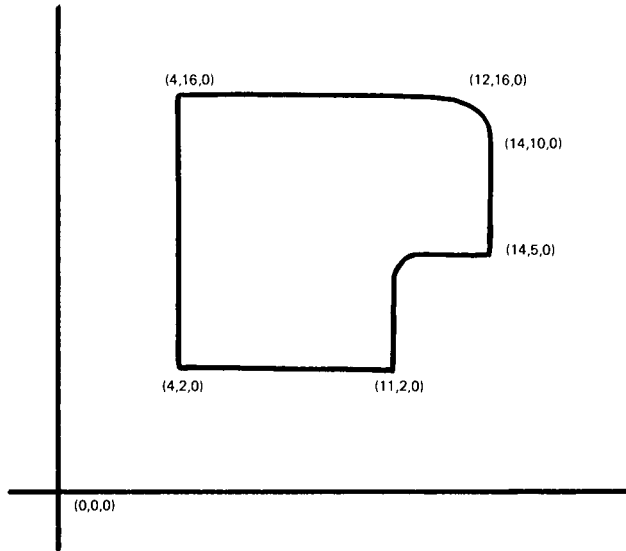


Fig. 13. Sample part with labeled machine coordinates.

the drilling of holes, the programmer would label the coordinates of the center of the holes and supply this information to the machine-tool controller, but if a continuous-path machine tool were used, as in our sample case, the part-programming job would become more complex.

The complexity of the part-programming task of these early continuous-path machine tools was due to the fact that it was written in a very non-operator-readable format, similar to writing a computer program in assembler. Add to this the fact that each different NC machine-tool controller required a different command word for the same procedure to be carried out from one machine tool to another (this is still true today), and the fact that when the path of the tool was programmed, the machine tool would move the tool in a straight line between succeeding coordinates. This last item was of major consequence, since in order to have the tool follow a curved path, many intermediate points were needed. These intermediate points would subdivide the curve path (see Fig. 14) into many short, straight-line segments (linear interpolation). This linear interpolation was a time-consuming and tedious task, since it was performed by hand, without the aid of today's electronic calculators or more powerful digital computers.

The closer the tolerance required, the more and smaller line segments needed to be calculated to define the curve. As the tolerance decreases, the number of line segments needed to generate the curve goes to infinity,

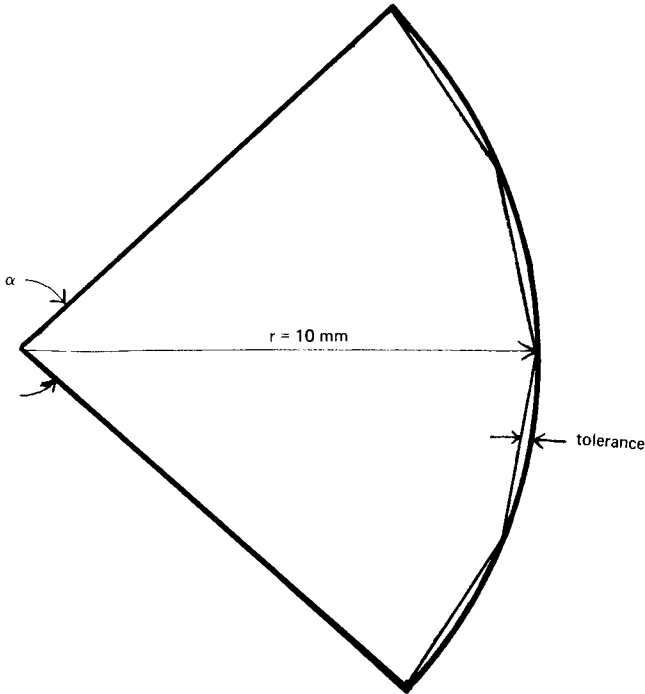


Fig. 14. Linear interpolation of a circular arc section via chords, where the circular arc length is $r\alpha$, r the radius and α the subtended angle.

according to the equation

$$m = \alpha/2 \arccos[1 - (\text{tolerance}/\text{radius})] \quad (1)$$

Equation (1) depicts the functional relationship between the number of line segments m needed to approximate a circular arc of length $r\alpha$ (α being the subtended angle) and a given tolerance (see Fig. 14). For more details see Pressman and Williams [9].

The part programmer's troubles are not over once all coordinate points are defined, because the tool dimensions have not yet been taken into consideration nor has the fact that rough cuts need to be made on the raw stock before the final finished cut is made. The sizes and shapes of the raw stock (see Fig. 15) need to be considered so that the appropriate transformation can be applied to the coordinates of the finished part in order that the early rough cuts may be properly defined (i.e., the appropriate coordinator for the tool to follow). Once these new coordinates are defined to reflect the raw stock, the part programmer needs to define the dimensions of the tool to be used and create and use the appropriate transformation

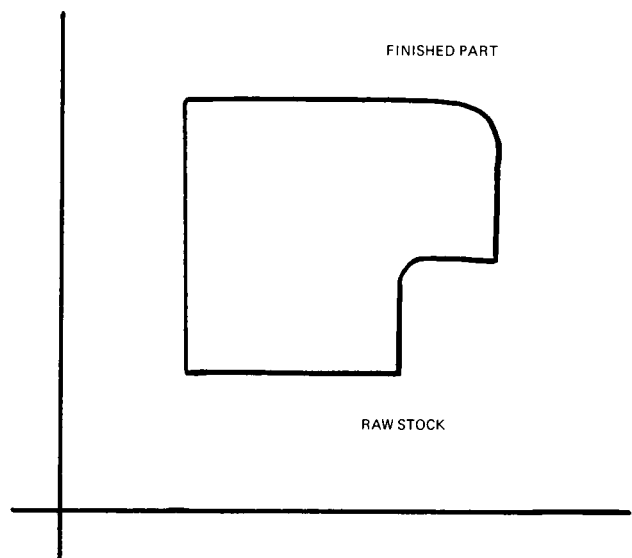


Fig. 15. Raw stock and finished part.

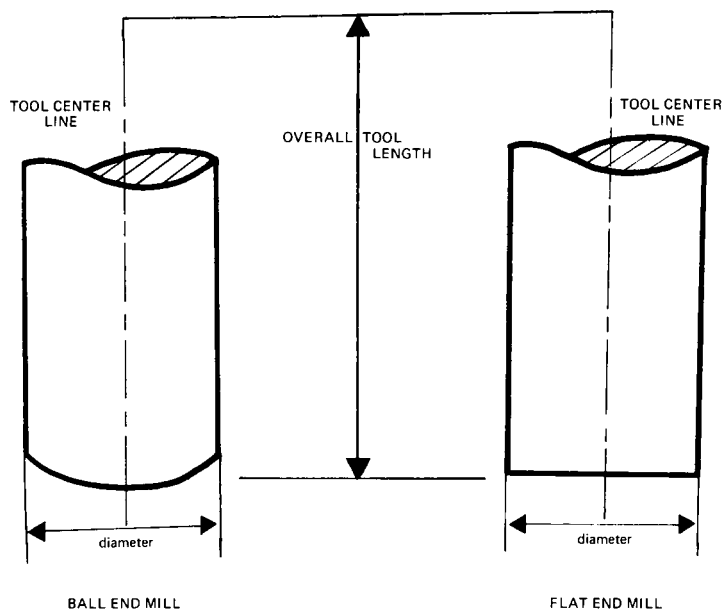


Fig. 16. Sample tools.

matrix on the calculated coordinates to reflect the size of the tool [i.e., tool diameter, length, if it has a flat or ball end, etc. (see Fig. 16)]. These specifications of the tool are needed since all coordinate locations are made with respect to some location on the machine-tool, and in order to cut the part properly, the part programmer must make the required transformation to reflect the tool being used.

More complexities are added to the part programmer's tasks when, for example, the end of a mill (either a flat-end or ball-end mill) is used to cut a contoured space curve. As can be seen in Fig. 17, where a ball-end mill is depicted, the point of contact between the tool and the part is, not the programmed point on the tool centerline, but is continually changing. This situation requires that time-consuming and tedious calculations be made by the part programmer, again a formidable task without the aid of today's electronic calculators or digital computers.

This process is complicated enough for a simple part, as depicted in Fig. 13, but such a simple part is not representative of the level of complexity involved in a myriad of parts machined by an organization.

In addition, this process (i.e., creation of a part program) was greatly simplified by the advent and subsequent application of the digital computer (see Fig. 12). In the earliest phase, digital computers aided in reducing the tediousness and time needed in the calculation of coordinate points used for linear interpolation and coordinate points needed for the correct cutting of space curves, as represented in Fig. 17. The digital computer was a tremendous boon to the part programmer, but this was just the beginning.

The next step was to increase the utilization of the tremendous processing power and speed of the digital computer. This was accomplished

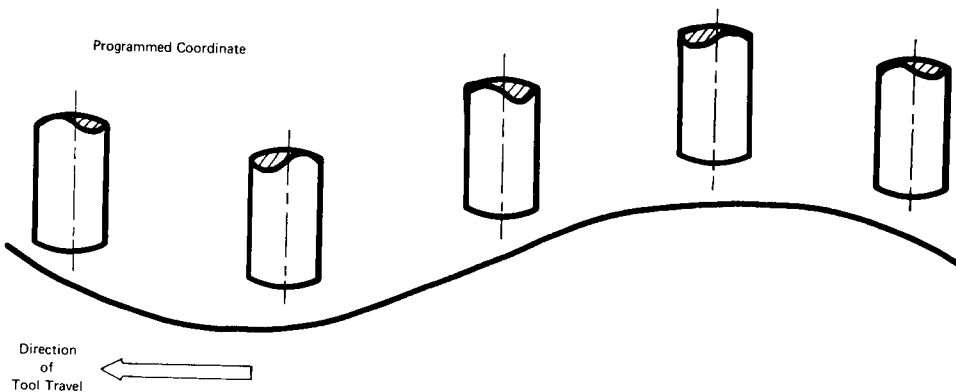


Fig. 17. Sample tool path profile.

by allowing the part programmer to define for the computer the basic geometric entities of the part (i.e., linear, circular, planar, etc.), the desired tool path, tolerances, and tool dimensions and then letting the digital computer process this geometric data and output the required coordinates defining the tool path. This enhancement came in the form of a computer program called APT, an acronym for automatically programmed tool. This program is the first, most powerful, and most widely used of the programs of its type. Its early development took place at the Illinois Institute of Technology, and it is continually being enhanced to make it a more powerful tool. It is a very large program and needs the power of today's large mainframe computer to run; because of this, other programs, which are basically subsets of APT, have been created to run on minicomputers (e.g., ADAPT, COMPACTII, MINIAPT).

The APT program and its related programs reduce the burden of the NC part programmer by taking more of the time-consuming and tedious tasks and transferring them onto the computer. The NC part programmer now needs only to define some coordinate system (either the machine tool's or a transformation to the machine tool's coordinate system); label geometric features and define them to the computer system (i.e., lines, points, circles); define the geometry of the tool; choose the machine tool to be used; and define the path, via geometry labels and APT vocabulary words. The computer will then calculate the coordinate points needed by the machine tool in order to cut the part. A sample part and its labels are depicted in Fig. 18, and its part program is depicted in Fig. 19.

Note that labels on the part engineering drawing (Fig. 18) correspond to geometric statement labels on the APT part program (Fig. 19). Labels L1–L4 define lines and C1 and C2 define circles, hence completing the geometric definitions needed for an APT part program. The next programming step is to define various machine-tool commands, such as tool dimension, feed rate, coolant on–off, and spindle speed, and then, finally, define the tool path by use of the geometric labels and APT vocabulary words. The computer will compute the necessary coordinate points needed to guide the tool while it cuts the part and associated feed rates.

The computer and APT combine to allow the programming of more complicated parts. Continuous-path machine-tool systems that manage simultaneous motions in two, three, four, or five axes by the interpolation of the path between prescribed points were realized. The fourth and fifth axes are angular orientations. Interpolations available are linear interpolation, already discussed, circular interpolation, parabolic interpolation, and cubic interpolation. These interpolators reside in the machine control unit and will be discussed in detail in the section concerned with controllers.

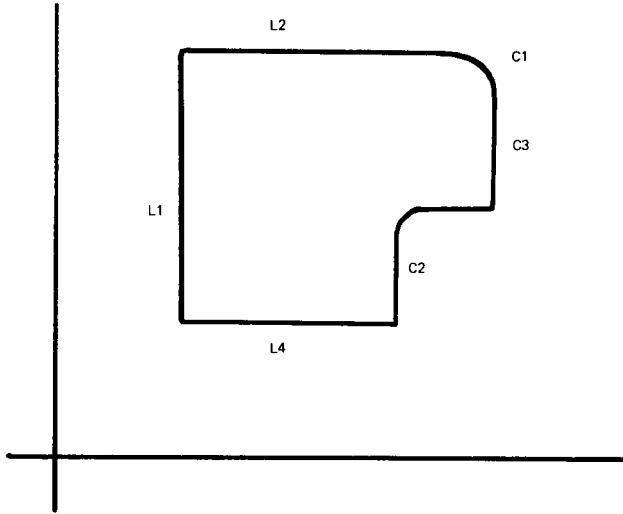


Fig. 18. Sample part labeled.

```

NOPOST
CLPRNT
PRINT/ON
DEFINE PART GEOMETRY TO APT
SB = POINT/ 0, 0, 0
L1 = LINE/4, 2, 0, 4, 12, 0
L2 = LINE/4, 12, 0, 12, 12, 0
L3 = LINE/ 14, 5, 0, 14, 10, 0
L4 = LINE/ 4, 2, 0, 11, 2, 0
C1 = CIRCLE/ 12, 10, 0, 2.0
C2 = CIRCLE/ 14, 2, 0, 3.0
INITIATE MACHINE
INTOL/ 0
OUTTOL/ .005
CUTTER/ .25
SPINDL/ 2000, CLW
COOLNT/ ON
FEDRAT/ 20.0
DEFINE TOOL PATH
FROM/ SB
GO/ TO, L1
TLLFT, COLFT/ L1, PAST, L2
GORT/ L2, TANTO, C1
COFWD/ C1, TANTO, L3
GOFWD/ L3, PAST, C2
CORTCT/ C2, PAST, L4
GORT/ L4, PAST, L1
GOTO/ SP
COOLNT/ OFF
SPINDL/ OFF
END
FINI
#END OF FILE

```

Fig. 19. Sample parts' APT part program.

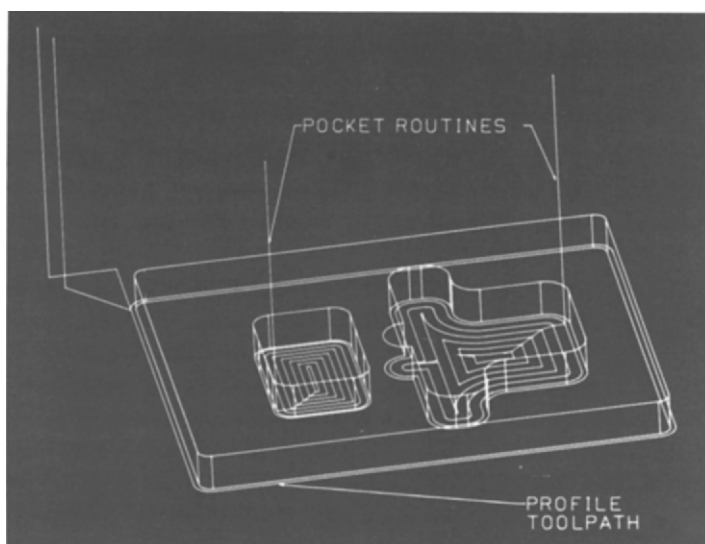


Fig. 20. Automatic pocketing and profiling, generated and displayed on a Computervision CAD/CAM system.

Advances made most recently in NC part programming due to developments in electronics have to do with interactive computer graphics. As discussed in Section III.B.1, interactive computer graphics has become a very powerful tool to the design engineer and draftsman. Once the data base defining the part has been created, this data base may be accessed and displayed by a manufacturing engineering group's NC part programmer. The NC part programmer uses the power of interactive computer graphics to interactively create the tool path in real time. Several large companies that have the manpower and computer capabilities have developed in-house systems, but several turnkey systems are presently on the market that have limited NC tool-path generation capabilities (limited to three-axis machine tools and three dimensions via wire frame models). Turnkey packages include such systems as CADAM, Lockheed; CADD, McAuto; Applicon, Inc.; Computervision, Inc.; and Gerber, Inc. Figures 20 and 21 were generated on a Computervision system and show the path taken by the centerline of a tool during automatic pocketing and surface machining. This type of system removes much of the programming task from the NC part programmer and allows the computer to calculate the required tool path.

Most of the burden that formerly resided on the shoulders of the NC part programmer has been shifted to the computer. The design engineer

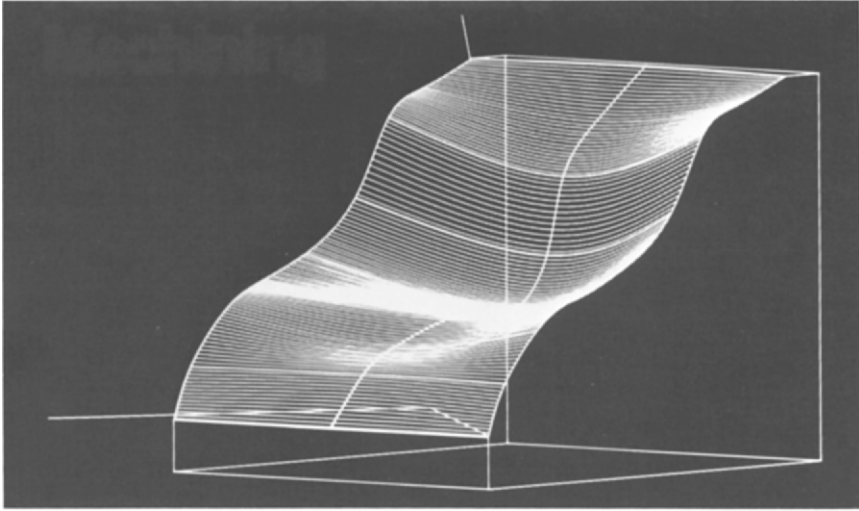


Fig. 21. Automatic surface machining, generated and displayed on a Computervision CAD/CAM system.

creates a geometric data base interactively in the computer memory and then the manufacturing engineer's NC part programmer uses this internally defined geometry to define the tool path interactively. The NC part programmer need not define any geometry to the system or write any program to be later submitted to an APT processor. Only machine-tool commands, such as coolant on-off, spindle speed, and tool dimension, need to be typed in, while all tool-path coordinates are automatically created by using the lightpen to interactively choose the geometric surface to be machined. On many of these interactive computer graphics systems, the APT or APT-like program need not be called to process the data since all tool-path coordinate points and auxiliary commands are in a form directly processible by the machine tool.

The use of interactive computer graphics in creating NC tool paths is as yet not as widespread and common as its use in design, but it is gaining acceptance. Present-day problems retarding its use in the NC part-programming area are its internal limitations (hardware and software) and system cost.

System cost is dropping and will be dropping more in the future, following trends in the past (more power at a lower price), and VLSI will make it more cost-effective, if trends continue, for small shops to afford interactive computer graphics systems for design as well as NC part-

programming generation. This will result in greater acceptance of this method of NC part-program generation, which at present is too costly for the small shops.

Limitations of interactive computer graphics acceptance for NC part programming arise from the fact that an efficient, fast, three-dimensional solids-modeling package needs to be developed. This package needs to shy away from purely mathematical definition (quadratic or cubic) of volume in order to reduce the calculation time needed to display complex objects.

The modeler needs to be able to define volumes that are not dependent on their axes being parallel, with any of the orthogonal axes, and following this thought, should not be limited to construction via logical operations on simple conic primitives. Such a three-dimensional solids-modeling package has been developed, as mentioned previously (see Section III.B.1).

Numerically controlled tool paths need to be validated as to accuracy of tool path, etc., and present-day verification techniques are either costly to perform or are not general enough.

The following three methods currently exist to validate the tool path: (1) cutting a prototype part on the machine tool; (2) plotting the tool path on a plotter; and (3) displaying the part and tool path in a two-dimensional turnkey interactive computer graphics system. Each of these methods has shortcomings.

The major drawback of machining a prototype is cost. Some elements entailed in the costs are the fact that the machine must be taken out of production, full setup is required, the part programmer and machine-tool operator must be in attendance, the raw material may be expensive (e.g., high technology alloy), and an error in the tool path can cause the material to be scrapped, with severe errors resulting in damage to machine and/or tool, etc. Based on figures compiled by Ward [10], a typical aerospace company supporting 75 NC machine-tools, verifying tool paths for 3777 items, for a total of 21,782 verification hours (at approximately \$51 composite hourly rate) over a 12-month period would cost \$1,110,882. This figure is conservative, since hourly composite rates can easily exceed the given figure by a factor of three or more, which is especially true of those jobs vended out. The cost of materials and additional costs can increase the cost of tool-path verification to an annual multimillion dollar expense. A figure of \$120 million/yr has been calculated by one aerospace company when all variables are taken into consideration.

Another NC tool-path verification technique is that of using a pen plotter, but this also has its shortcomings. Use of a plotter is too slow for verifying large numbers of tool paths. A large tool-path program may take

an hour or more to plot, and there is the question of reliability with respect to tolerances. Another shortcoming is that verification of tool paths for machine tools that have three or more axes is impossible since the plotter works on a two-dimensional plane. Thus it is difficult to visualize and verify that all axes move correctly. The same arguments can be applied to the use of turnkey interactive computer graphics systems [item (3)], since these too are two-dimensional.

Although most commercial turnkey interactive computer graphics systems provide wireframe and surface definitions for the objects in their data base, a solids-modeling package is more desirable for NC tool-path verification. A solids-model definition for an object allows a volume of three-dimensional space to be occupied by the object. This allows testing of such things as interference of the tool, fixtures, and clamps. Color enables the object violation to be readily distinguished. Overcut and undercut of the part by the tool, a good tool cut (within set tolerances), or interference between tool and clamps and/or fixtures can be identified by appropriate color differences in the displayed picture.

A solids-modeling color system, which readily lends itself to the possibility of real-time NC tool-path validation with the advent of VLSI, is the octree system discussed in Section III.B.1. The Center for Manufacturing Productivity and Technology Transfer at Rensselaer Polytechnic Institute is presently doing a feasibility study related to the use of octree in NC tool-path validation. Figure 22 depicts a simple part, tool, and clamp as generated by the octree program, except that in the original the part was red, the tool blue, and the clamp green. The tool has made two cuts into

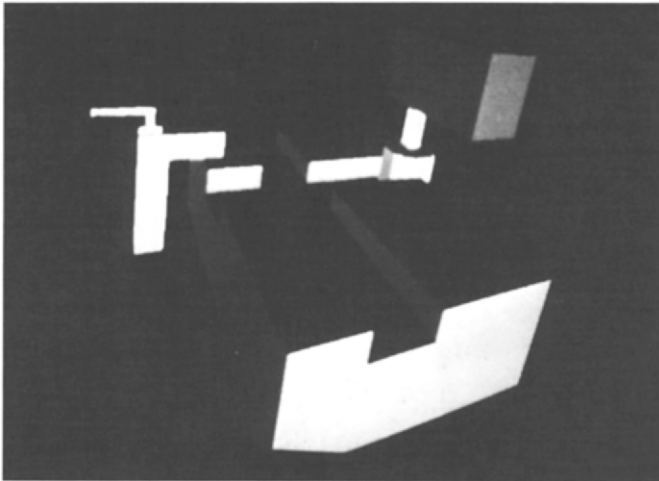


Fig. 22. Octree-generated picture.

the part by removing the volume swept by the tool from the part. Some examples of work being done to allow the tool to follow an arbitrary space curve, cut a more complex object, allow real-time interaction, and depict over- or undercut via color coding.

The Aerospace Industries Association (AIA) Computer-Aided Manufacturing Task Advisory Group (CAMTAG) Manufacturing Committee (MC) has sponsored a project (MC79.6) entitled Computer Verification of Machine Control Data, which is based upon developments in the capabilities and versatility of interactive computer graphics technology.

Hardware advances, such as denser memory with high access rates are needed to display and manipulate the solid objects in real time. VLSI is needed to allow more powerful color interactive graphics terminals, which can be accessed interactively and, in real-time, manipulate complex color objects (i.e., rotation, scale, translation).

a. Voice Input. Areas other than the display and manipulation of color graphics images needing development and upon which VLSI can have a great impact in NC part programming are auto cut vectors or machine programming and voice input (see Fig. 12).

Auto cut vectors refer to a generative process by which the NC part program or, rather more accurately, the NC tool path is automatically generated by the computer based upon the computer-resident three-dimensional solids-model data base. The only interaction then required by a human would be in verification of the computer-generated tool path (machine-tool choice and required machine parameters would be automatic), verification as to overall motion of the tool, and verification of color encoded tolerancing.

Machine programming refers to powerful VLSI circuitry on the machine-tool controller in order to create the tool path as previously described at the machine-tool site rather than elsewhere.

Voice input refers to the usage of powerful VLSI circuitry, which would allow an NC part programmer to sit in front of an interactive computer graphics display terminal and verbally request the needed part be displayed. Once the part is displayed, manipulation of the displayed object (i.e., rotate, scale, translate) can be accomplished by voice commands, and then, the creation of the tool path can be done either generatively, as previously described, with the part programmer verbally initiating the task and supplying verbally any auxiliary commands, or a new tool path can be created by the NC part programmer verbally telling the system to which system-labeled surface to move the tool, at what depth of cut, and so on.

An example of such a voice system for NC programming has been

developed by Threshold Technology, Inc., and is marketed by the name Voice Numerical Control System (VNC).

Voice input is today's state of the art in computer-assisted programming of NC machine-tools. The VNC system offers three distinct advantages over all other systems:

(1) **Simplicity of operation.** The VNC system requires knowledge of good machining practices, not specialized computer expertise. The programming language itself consists of fewer than 100 words, all of them common shop terms, such as "machine line," "point circle," and "thread."

(2) **Error-free tapes.** The VNC system requires no manual steps; the programmer enters data into the system with by voice. Eye-voice coordination (rather than eye-hand coordination) accounts for 99% accuracy. Four further safeguards ensure total accuracy: immediate verification via alphanumeric readout of each command, CRT display of all commands, a plot of the part as it is being programmed, and a printout of all machining steps.

(3) **Economy of time and money.** The VNC system produces program tapes five times faster than even the large computers used by time-sharing firms and at less than half the cost. The use of machine-shop skills and the elimination of all manual steps minimize direct labor overhead. Fast production of tapes keep expensive machine tools operating at peak capacity.

The VNC system is first trained by the NC part programmer, who repeats each word of the vocabulary five times to provide a reference-set of features. A noise-canceling microphone is used by the programmer in order to reduce interfering background noise and acoustical noise. Boundaries of the words must be learned by the system (i.e., the actual beginning and ending of a word), which requires pattern-recognition processing to discriminate between the breathing noises of the programmer and the actual speech sounds.

A spectrum analysis of the waveform produced is carried out resulting in 512 bits of information being stored as a reference map for each word. Once this information is stored, the VNC system will compare the part programmer's future speech patterns for words with these learned words and allow the verbal creation of a part program. Savings in part program creation of 50% have been realized.

Much needs to be done in the area of computer voice-recognition systems and this is just the beginning. VLSI and denser memory capabilities will supply the needed power at an affordable price, if present trends continue.

There are, to be sure, many other areas that can be impacted by VLSI with respect to NC part programming, but the effort here has been to show the historical trends and to encourage progress in the future.

2. Process Planning

Traditional process planning, which is the planning of the flow of raw material through the manufacturing process in order to most efficiently utilize man and machine through material selection and sequencing of operations and machine selection, was previously accomplished by manufacturing engineers who had been with an organization for years and had intimate knowledge of the factory floor. The recent rapid turnover in manpower does not allow a pool of knowledgeable manufacturing engineers to develop, and the movement, as in other areas, is to place the knowledge of an experienced process planner into the computer memory. This placement allows, with proper programming, the novice manufacturing engineer to plan the process with the aid of the computer interactively and yield the results that a senior employee would. Thus the computer's impact is seen in process planning the extension of man.

As seen in Fig. 23, the evaluation of computer usage as an aid in process planning parallels that just covered for NC part programming, although computer use as an aid for process planning has not developed to the degree that it has in NC part-program creation.

Time sharing along with interactive computer networking made the

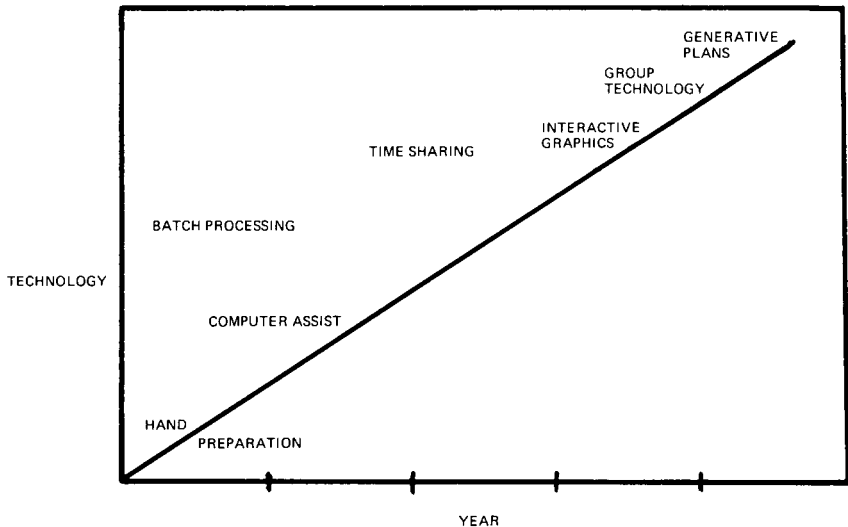


Fig. 23. Process planning.

computer a very powerful tool to the process planner. Information concerning the status of material in process and the status of machine-tools is now made readily available to the process planner. This is accomplished by the process planner requesting the needed information interactively at a CRT and having that information displayed at the CRT in a format readily understandable by the planner. Creation of the plan still requires a knowledgeable planner at this point of the development since the computer merely supplies needed information—not merely information, but information supplied rapidly, very accessibly, and currently. The planner uses this information and the CRT to plan the process of other material through the factory floor.

The advent of interactive computer graphics will allow the display of the shop floor, yielding information concerning the status of machine tools and material flow to be graphically represented. Incorporation of intelligence into the terminal, as made available by VLSI, would allow such a system to create a process plan for a part generatively. The planner would need only choose with the lightpen which processing nodes are required in the proper sequence and the computer would create the process schedule for the part automatically. The plan thus generated would reflect scheduling of other parts through the system, yielding an optimization of machine time.

Another method by which a process plan can be automatically generated is through the use of group technology, or master family planning. Parts with similar characteristics, say, with respect to processing requirements, are grouped together. Each group has a standard, basic, process plan associated with it, and as requested, the computer will generate a process plan for a particular element of the group and integrate the plan around other existing schedules.

Only the downloading of such tasks from the mainframes of today, through the use of VLSI, will allow the full impact of such an automatic process-planning system to be realized, for not only is information needed concerning part creation but information on the status of machine tools and parts in process is needed and the system must have the power and programming to allow it to create an optimized plan based on needs and resources. With the aid of a fully optimized process plan, the full productive potential of the shop floor will begin to be realized, and VLSI has the potential to assist in this endeavor.

D. Production

Production entails not only the actual manufacturing of the product but also the control of the manufacturing process. Included on this level of our CAM structure are such entities as inventory, distribution of raw

materials, production control, machine tools, robots, quality assurance, final assembly, and, of course, shipping.

1. Control of Material Flow

Under the heading “Control of Material Flow” are such topics as inventory and distribution of raw material. Both of these items are in the realm of management, and the discussion of current data bases and flexible access to these data bases, as covered in Section III.A is applicable here. Of course, the contents of our data bases on this level deal most basically with such items as how much stock (i.e., raw material, tools, spare parts) currently exists; what are current needs; where is the stock needed; and what is the status of the stock already on the factory floor?

Inventory/distribution systems that use the processing powers of the computer and are integrated with the stockroom and the shop floor form the basis for substantial gains in overall productivity. This is reinforced by the fact that, of the total time a part spends on the shop floor, only 5% of that time is spent in actually cutting the part. The remaining 95% of the time is divided among such tasks as setup time on the machine, movement of the part on the shop floor, and time spent in queue.

The integration of VLSI technology, robotics, and lasers can lead to a system such as the following. Parts of such systems exist today; what is

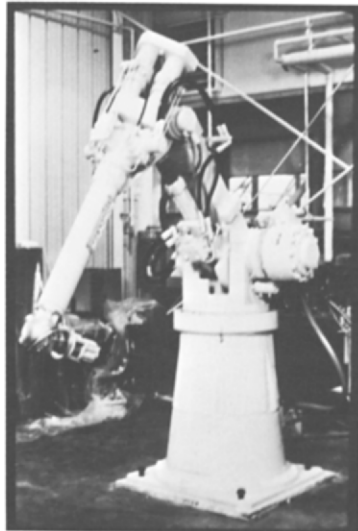


Fig. 24. T³ industrial robot (courtesy of Cincinnati Milacron).

needed is a complete integration of such a system into a CAM structure of information flow.

Lasers can be used to scan coding on incoming stock whose code is automatically logged into an appropriate data base. Robots, under computer control, can then be used to store this incoming stock in a storeroom. When the time comes to machine a part, the computer-controlled robot will retrieve the part from storage.

In small manufacturing cells, robots can be used to load and unload machines to aid in the process flow. Figure 24 shows a Cincinnati Milacron T³ industrial robot serving a manufacturing cell consisting of two Cinturn NC turning centers. The robot and machine tools are under computer control and can be connected via links to a CAM hierarchical structure.

Figure 25 depicts a variable-mission manufacturing system developed by Cincinnati Milacron. It consists of a 13-station computer-controlled machining system to reduce manufacturing costs and inventory and to increase production and flexibility for a family of 10 different aircraft engine cylinder heads. The system performs nearly 50 metal-cutting operations on each part. Rough castings are loaded on pallet-carrying carts in the staging area, transported, deposited and picked up at any of the 13 machining stations, and returned to the staging area by a computer-controlled, wire-guided, parts-handling transport system.

The flow and distribution of material is controlled by a central computer for the system. Lasers can be distributed in the system to scan codes existing on the part or pallet and, hence, to allow the computer to keep track of material within its domain. Information from this manufacturing cell and others can be fed into a central processor, which overviews the whole factory and controls the flow of material. Optimization of the flow of material can be accomplished, reducing the queuing time for parts on the floor; palletizing removes the time needed for numerous machine setups, and hence, we can foresee a time when a part on the factory floor does not need to spend only 5% of its time actually being machined. The result will be an overall increase in productivity.

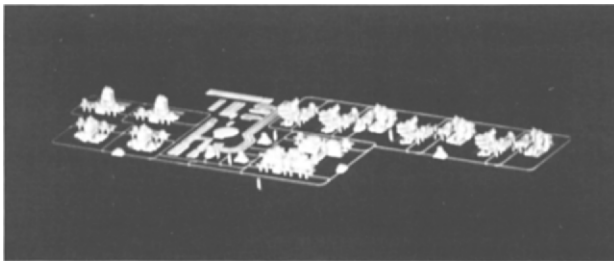


Fig. 25. Variable-mission manufacturing system (courtesy of Cincinnati Milacron).

Bits and pieces of such a system exist, and aided by the implementation of VLSI, a fully integrated system encompassing the whole factory can be realized, with VLSI supplying overall and local processing powers needed for such a system, along with parallel developments in communications.

2. Manufacturing

Manufacturing is the process of converting raw material into some useful form, and what is done on manufacturing floors of companies throughout the world is exactly that. Raw material is delivered to the shop floor through some distribution process and is shuttled from one machine site to another until it is transformed (in our case, machined) into some useful entity. This entity is then shuttled to a quality assurance section where it is checked for conformity to design specification. Upon successful completion of inspection, the entity is either shipped to a customer, put into storage, or shuttled to a final assembly area where the newly formed entity becomes part of a larger, more complex entity.

The main topics of discussion in this section will be NC machine tools, quality assurance, and final assembly.

a. NC Machine Tools. In our discussion of NC machine tools, topics to be covered are machine controllers, adaptive control, and machine diagnostics.

Controllers for numerically controlled (NC) metal-cutting machine-tools have followed the trend depicted in Fig. 26. The controller for an NC machine tool must control the motion of a tool that is under a tremendous dynamic load to accuracies in the order of 0.0001 in. The control must dynamically respond to characteristics of the machine tool in order to accurately guide the tool within tolerance and control many simultaneous movements of the machine-tool, which may weigh up to 100 tons.

Since the advent of numerical control machine tools in the 1940s, the movement has been from controllers that are all hard-wired, logical circuitry, inflexible to systems that integrate computers into their controllers. The computer-integrated systems are flexible in that controlling functions are executed by software rather than hardware and changes in a control function require software modification rather than rewiring of circuitry.

Depicted in Fig. 26 is the trend in control units growing in parallel with developments in electronics. We can see the movement from hand-wired, logical circuitry, vacuum tubes, transistors, integrated circuits, LSFs, and on to VLSI. The general trend, as electronics develops, is the creation of smarter, more automatic, and easily implemented controls at an approximately constant or decreasing price. Mechanical complexity is

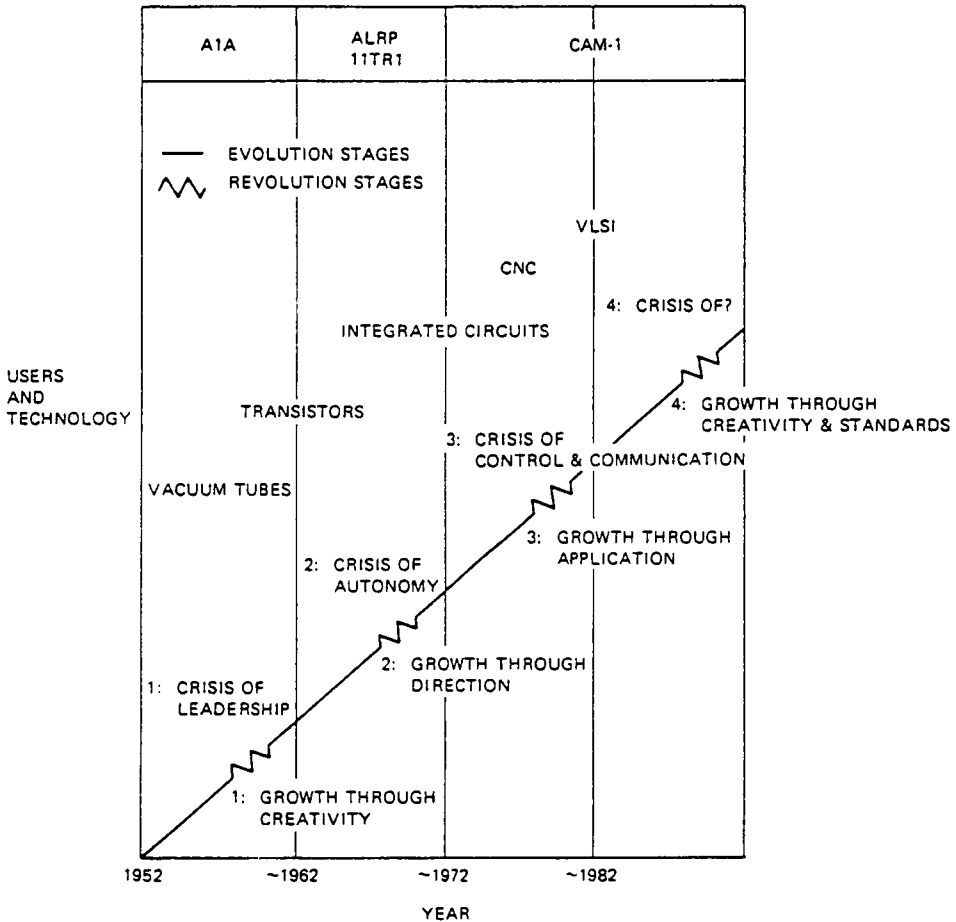


Fig. 26. NC growth (courtesy of Dr. R. Miskell, Machine Tool Task Force.)

being replaced more and more by sophisticated electronics with a favorable reliability trend. A key technical issue will be the development of functional control interfacing standards in support of more productive CAM systems. Such standards will provide important software and hardware guidance for expanded shop flow process control and remote computer communications.

There exist two directions in control development and these have to do with what is called computerized numerical control (CNC) and direct, or distributed, numerical control (DNC). Computerized numerical control deals with increases in the breadth of process control for machine tools by adding more on-site (i.e., at the machine tool) capabilities. These capabil-

ities include such things as NC program editing, machine and control diagnostics, adaptive control, tool data management, part or workpiece measuring, etc. Direct numerical control deals with the integration of process control into a hierarchy of machine control with communication between machines, two-way data communication, system operations, and planning functions.

As defined by the Electronic Industries Association, CNC is

a numerical control system wherein a dedicated, stored program computer is used to perform some or all of the basic numerical control functions in accordance with control programs stored in the read-write memory of the computer.

Hence, CNC refers to a NC machine-tool whose controller is a dedicated computer.

As defined by the Electronic Industries Association DNC is

a system connecting a set of numerically controlled machine-tools to a common memory for part-program or machine-tool program storage with provisions for on-demand distribution of data to machine-tools.

The important distinction between CNC and DNC is that CNC is a dedicated computer, whereas DNC is a hierarchical structure with a central computer from which controlling information may be set to numerous machine tools. Figures 27 and 24 depict CNC and DNC systems, respectively. Figure 27 is a Cincinnati Milacron 10VC CNC vertical machining center.

At present, LSI and, in the future, VLSI technology will continue to expand and offer many bipolar and metal-oxide-semiconductor (MOS) devices for use in machine-tool control units. In CNC control units, 16-bit high-performance microprocessors and minicomputers dominate, but in

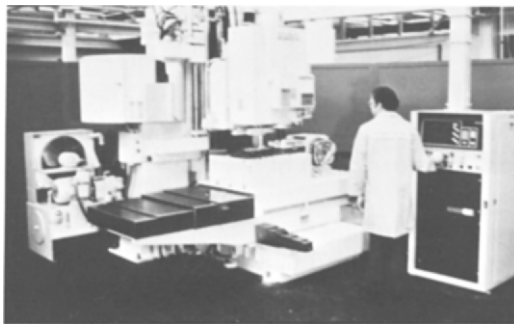


Fig. 27. 10VC CNC vertical machining center (courtesy of Cincinnati Milacron).

many applications, performance enhancements are achieved through the use of parallel processors. These parallel microprocessors perform data handling and control functions in a distributed fashion.

As noted by T. Trexler, Bendix Corporation [11], some features presently included in CNC that would have been difficult to include on hand-wired controls are

- (a) on-line editor, prompting, and paging schemes;
- (b) multiple-block buffering of part-program data, directions of large part programs, work space for modifying part programs;
- (c) visual display on CRT, displays of diagnostic messages, ease of use with conventional programming techniques, and other CRT enhancements;
- (d) English-language instructional techniques through the CRT to permit the operator to select the correct modes and build the part programs;
- (e) marriage parameter initializations;
- (f) closing of servo loops;
- (g) offset and compensation data such as:
 - (1) tool-noise-radius compensation,
 - (2) cutter-radius compensation,
 - (3) tool-length compensation, offsets, and auto gauging, fixture offsets, leadscrew-error compensations, and reversal-error compensations;
- (h) software travel limits;
- (i) software interpreters for resident programmable controllers;
- (j) tool-life algorithm; and
- (k) digitizing methods for generating a part program.

These are just a few few examples of the myriad of enhancements that are possible through a machine-tool computer marriage.

Figure 28 depicts the advances in curve-generation power of the NC machine-tools controller correlated with respect to time and developments in electronics. We can see the trend in increase in complexity and accuracy in curve-generation technique, starting at linear interpolation and advancing through circular interpolation, parabolic interpolation (see Childs [12], for more detail), then, with the introduction of CNC, implementation of general algorithms, examples of some given in the preceding paragraph, and, finally, the capability of automatic cut vector generation.

Malfunction analysis or diagnostics is an area that needs much develop-

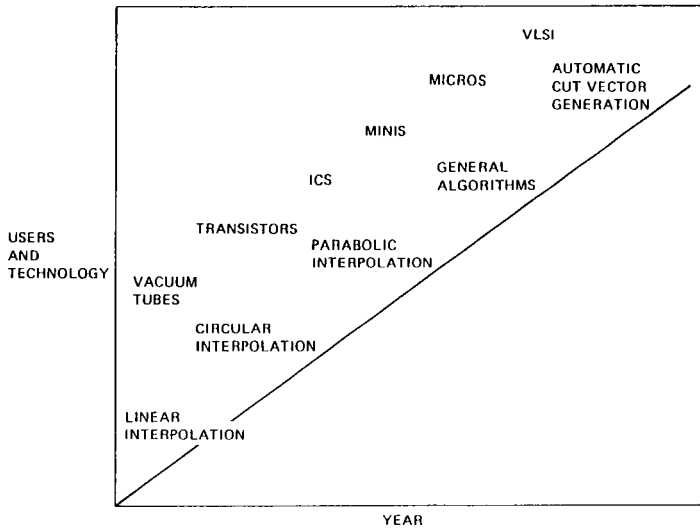


Fig. 28. Advances in curve generation (courtesy of Dr. R. Miskell, Machine Tool Task Force.)

ment. At present, limited real-time diagnostics are available for NC machine-tools, but these diagnostics are with respect to the computer control unit and not the machine tool itself. The trends over the years are depicted in Fig. 29.

What is needed is not only diagnostics of the computer control unit but also of the mechanical part, the machine tool. The diagnostics should be

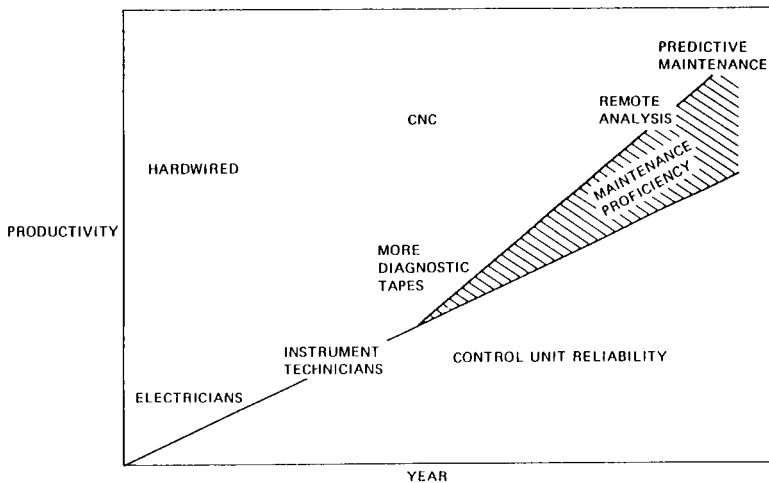


Fig. 29. Diagnostic advances (courtesy of Dr. R. Miskell, Machine Tool Task Force).

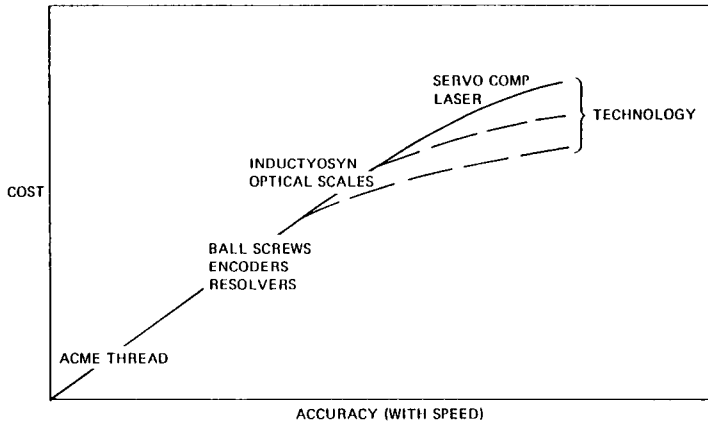


Fig. 30. Feedback improvement (courtesy of Dr. R. Miskell, Machine Tool Task Force.)

done in real time with VLSI and parallel processors making this a reality. Diagnostics should be structured to allow prediction, detection, diagnosis, and repair instructions.

Improvements in the prediction and detection of a malfunction will be due to improvements in feedback capabilities. Figure 30 shows a cost versus accuracy (with speed) feedback technology. Incorporation of these elements with increased processor capabilities will allow for needed malfunction prediction and detection.

Figure 31 shows a malfunction analysis from prediction to repair. Prediction of the malfunction is accomplished via comparison of stored oil pump parameters with data attained via the feedback system. When feedback data is approaching the stored parameter limits, a warning message appears on the CRT. Upon failure of the oil pump feedback, data will be outside the range of stored parameters and a message stating the device's failure will appear on the CRT along with the reasons for the failure and the appropriate repair procedures.

Adaptive control is another area that needs to be further developed in

- PREDICTION
THE SPINDLE BEARING OIL PUMP WILL FAIL
- DETECTION
THE NO. 2 SPINDLE BEARING HAS FAILED
- DIAGNOSIS
FAILURE OF THE NO. 2 SPINDLE BEARING
DUE TO OIL PUMP FAILURE
- REPAIR
REPLACE SPINDLE OIL PUMP AND NO. 2
SPINDLE BEARING

Fig. 31. Malfunction analysis (diagnostics) (courtesy of Dr. R. Miskell, Machine Tool Task Force).

- AUTOMATIC MODIFICATION OF COEFFICIENTS IN TRANSFER FUNCTIONS
- OPTIMIZE FEED AND SPEED BASED ON SENSORS
- AUTOMATIC SCREW PITCH CORRECTION
- AUTOMATIC INTEGRATION OF MACHINABILITY DATA
- CUTTER COMPENSATION
- SPINDLE GROWTH COMPENSATION
- TOOL OFFSETS

Fig. 32. Some meanings of adaptive control (courtesy of Dr. R. Miskell, Machine Tool Task Force).

order to create real-time adaptive control that has the capability of sensing certain process parameters and superimposing process constraints so as to provide the corrective response necessary to improve the process condition within these constraints. Adaptive control can take on many meanings, a few examples of which can be seen in Fig. 32.

Adaptive control systems today assure a stable process condition with respect to a particular process variable such as cutting torque or cutting power (see Fig. 33). Development of these systems is based, in part, on analog hardware. Already available is the CNC, which offers the possibility of integrating the constraint control in the computer. Powerful control algorithms with adaptive control modifications can be realized with the aid of the computer with the addition of hardware at a minimum.

The growth in adaptive control is depicted in Fig. 34, and with the aid of VLSI, real-time adaptive control over many parameters can become a reality. Associated with increased adaptive control capabilities is the parallel development of sensors. Sensor research, as applied to metal-cutting machine tools, includes developing the relationships among the aim (i.e., what is to be detected), the correcting variable, the monitoring parame-

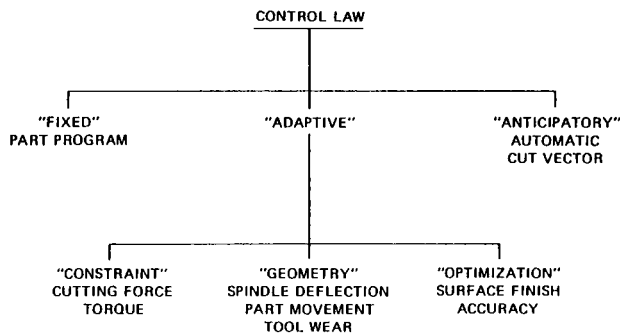


Fig. 33. Control strategies (courtesy of Dr. R. Miskell, Machine Tool Task Force).

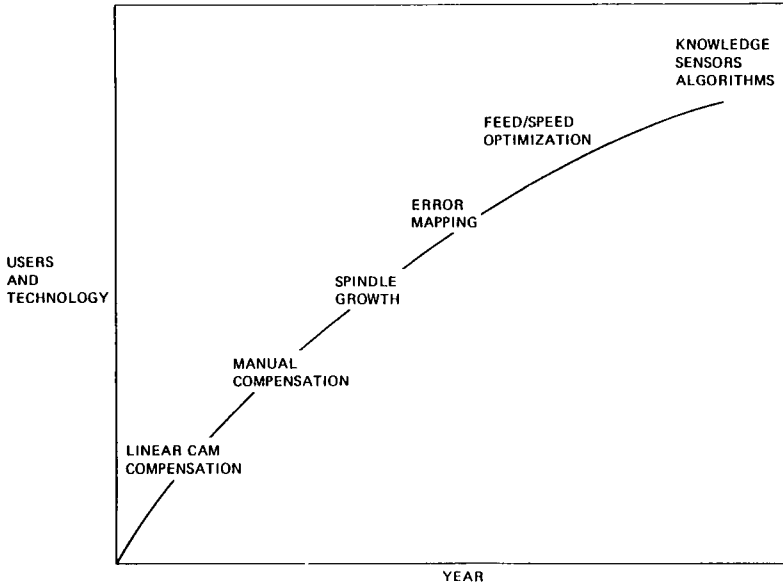


Fig. 34. Growth in adaptive control (courtesy of Dr. R. Miskell, Machine Tool Task Force).

ters, and the sensed measure. Sensing includes integrating measured physical quantities and processing of these raw signals to derive more meaningful data and, subsequently, outputting an interpretation of these signals. Sensing systems used to check for worn or broken tools, chatter, slide location, etc., must include the ability to calculate and send appropriate commands that may be used by an adaptive control unit or a diagnostic system.

The processing power and speeds associated with VLSI make real-time adaptive control, diagnostics, and sensing realizable in the near future as part of a CNC machine tool. Much work needs to be done in these areas to realize these goals, but the payoff will be in increased productivity, integrity, and reliability of our manufacturing facilities. Areas that need development are automatic cut generation, standardization of interfaces, both hardware and software, on-line servo compensation, ac drivers, laser feedback, geometric adaptive control (tool wear, spindle deflection), and predictive maintenance via sensors and algorithms. Table II depicts functions performed by a CNC unit today, tomorrow, and in its possible future.

b. Quality Assurance. Once a part has been machined, one does not know if, in fact, the part has been cut properly. In other words, has the

TABLE II
Functions Performed by a CNC Unit^a

Today	Tomorrow (near future)	The future perhaps
Linear and circular interpolation	Assistance in accuracy calibration	Control of coupled storage and retrieval system for parts
Metric or inch selection	More automatic tool setting	Use CNC display and computer with machine for operators' training and, perhaps later, for maintenance technicians' training
Tool selection	More adaptive control	Display of workpiece, fixture, and cut for program verification
Block delete	Error correction	On-line signature analysis
Position set	More control of coupled automatic parts handling (pallet system, conveyor, or robot)	Integration of process planning functions, process optimization using material properties, and process data
Backlash compensation	Self-diagnosing and self-repair system (e.g., broken drill)	Automatic redundancy switching
Canned subprograms (threading, tapping, mirror image)	Feeler probes for automatic part location	Better identification of intermittent faults
Inhibits, such as protected work space	Control of coupled automatic inspection system	
Ability to modify program on the shop floor	Record events of last few minutes of operation prior to a failure (for maintenance and warranty)	
Feeler probe for part inspection	Automatic cut distribution	
Recordkeeping of machine utilization, tool usage (tool life), parts made	Expand on-the-floor part-programming capability	
Some adaptive controls	Expand contour display of cut for program verification	
Some diagnostic devices or error messages to detect failure	Expand use of CNC display and computer for maintenance troubleshooting	
Simultaneous control of two cutting tools		
Tie CNC into DNC or FMS		
Tie CNC to remote computer for diagnosis routine		
Cutter compensation		
Safety protection: spindle overload and speed		
Programming parts at the machine		

^a Courtesy of R. Miskell, Machine Tool Task Force.

machine tool cut the part within tolerance as required by the engineering drawing? This fact can be ascertained by inspecting the part.

Inspection can take the form of manually setting up the part, using gauges, manually collecting data, and making calculations to assure that a part is either within specified tolerances or not. With the advent of computers, another technique for part quality inspection has been devised in the form of computerized coordinate-measuring machines (CMMs).

Computerized CMMs can be either semiautomatic or automatic, similar to CNC machine tools. A semiautomatic system requires a software program be created by a quality assurance inspection part programmer to instruct, via printout on a CRT at the CMM site, the CMM's operator as to the geometry needed to be measured. The CMM's operator will manually guide the CMM's probe to touch the specified part location. Coordinate points of this location will be read by the computer and used in various calculations (e.g., base diameter, concentricity). The calculations to be carried out are part of the sequence of programming that resides in the CMM's computer.

The automatic computerized (CNC) coordinate-measuring machine is similar to CNC machine tools in that the CMM's probe is guided by the computer in touching and, hence, inspecting the part. An inspection part program resides in the CMM's controlling computer and contains commands to move the CMM's axes to specific locations automatically. Commands are transmitted to each axis driver, as in the machine tool, and when a probe contacts a surface, the coordinates of that location are stored in memory and are used as input to calculate whether certain crucial geometric properties of the part have been cut correctly (e.g., bore diameters, concentricity of bores).

Figure 35 shows computerized coordinate-measuring machines with physical probes marketed by Mechanical Technology, Incorporated's Boice Division. Figure 36 shows the Boice VISTA¹ Automatic Video Inspection System.

The VISTA system is an automatic measuring system combining advanced video technology with microcomputer speed.

The VISTA System inspects parts through a technique called digitizing. A TV camera "sees" the part and displays its image on a TV screen. A two-dimensional grid, dividing the image into one-quarter-million elements, is used to inspect a part feature by analyzing its representation as elements of that grid. Using this grid information, a microcomputer (video processor) calculates the feature's size and position, compares them with

¹ VISTA is a registered tradename of the Boice Division of Mechanical Technology, Inc.



Fig. 35. ACRA-CORD CMM CL-201 (courtesy of Boice).



Fig. 36. VISTRA system (courtesy of Boice).

part-print nominals and tolerances (stored information), and prints an inspection report. This report appears on the screen, and a hard copy is readily available. Any data out of tolerance are noted as "fail" on the report, which can be attached to the part for future reference. Digitizing makes the difference in accuracy.

The VISTA system measures accurately to 0.0003 in. Variations in accuracy caused by visual acuity, operator technique, and human judgment are minimized because measurements are done by a microcomputer. The operator does not influence measurements or interpret results; the microcomputer locates desired features, measures them, and reports out-of-tolerance data, all automatically.

Noncontact measurements on small, difficult to inspect, and fragile parts are performed without damage by the VISTA system. Accurate dimensional verification of precision components is fast and easy. For the manufacture of large quantities of flat or complex parts, the VISTA system dramatically improves the inspection process. Inspection man-hours are reduced drastically by the VISTA system. It is up to 20 times faster than an optical comparator, i.e., less than 1 sec/measurement. Parts are inspected so rapidly that flaws in a production machine can be recognized and corrected before "reject" parts accumulate. Shorter per-part-inspection time, high accuracy, and exact identification of flaws all help to cut quality control costs.

Operating the VISTA system is easier than measuring with optical comparators. Everything is easily accessible. The operator need not visually align the part image with comparator chart overlays or microscope cross hairs. Step-by-step inspection is done simply by pressing buttons on the keypad; no knowledge of computers is necessary. The VISTA system remembers (programs) the measuring sequence and performs it automatically on the next identical part. Part-program storage is unlimited; any program can be called into use within seconds. Both large and small parts can be inspected simplifying overall quality control.

The VISTA system is programmed as easily as a hand-held calculator. By simply pushing the required buttons, the operator can move the camera over the area to be inspected, adjust the size of the window to fit the feature, and position the cross hairs to make the required measurements. Pushing the STORE VIDEO button puts this sequence into VISTA's memory. The process is repeated for all selected measurement points after determining which measurement category is desired. Nominals and tolerances are typed in and the entire inspection sequence is now ready for use. For identical parts, the operator will load the part and touch the RESET and RUN buttons, and inspection will be totally automatic. The printer runs off a hard copy of the sequence so that a report can be

attached to the part, with a pass (P) or fail (F) indication. This process is simply repeated for each part to be inspected. If a different preprogrammed part is next in line, entering the part number and touching RECALL will begin the new inspection process. All programs can be stored on floppy disks for future use.

Manual inspection can be done with no program and enables the operator to examine one-of-a kind parts quickly and accurately.

Work is being done to make VISTA a three-dimensional system, and with VLSI, a more powerful tool can be developed in this area.

A "seeing" computer, called SIGHT-1, developed jointly by GM's Research Laboratories and Manufacturing Development, has recently become the first of its kind to go to work on a U. S. automotive production line (see Fig. 37). Using an electronic camera as its "eye," this computer vision system inspects and calculates the position of integrated circuit chips at Delco Electronics Division in Kokomo, Indiana. Other promising computer vision systems are being developed at GM Research Laboratories.

Computer vision, as the name implies, is the new science of giving computers "eyes." Looking through electronic cameras, computer programs sense, interpret, and analyze visual scenes.

What are computer vision systems good for? Their potential applica-

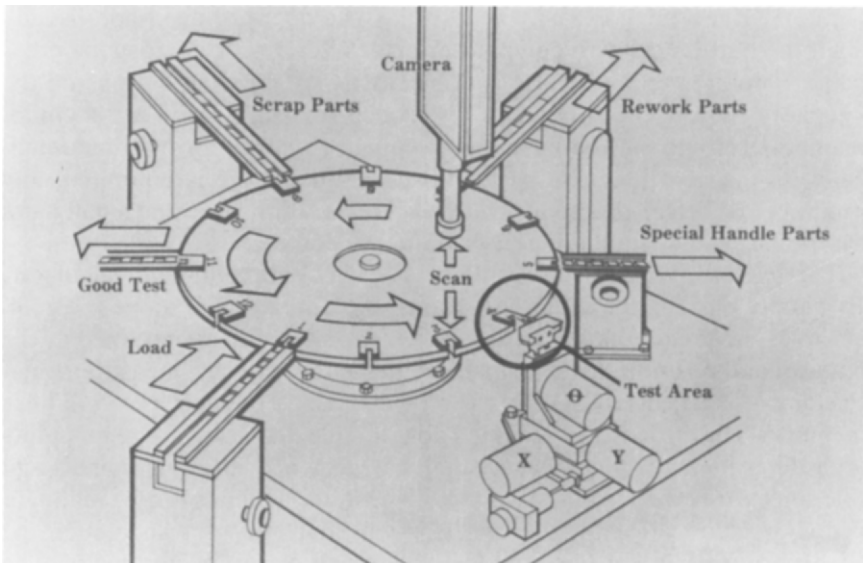


Fig. 37. SIGHT-1 at work.

tions are unlimited. They can inspect parts. They can serve as eyes for robots, making possible completely automatic part transfer and assembly. They can reliably perform difficult assembly-line tasks. And they shrug off hazardous environments.

SIGHT-1 is the first industrial computer vision system of its kind put to use in the United States. It was installed at Delco Electronics in Kokomo, Indiana, in January 1977. Its task is to inspect and calculate the position of integrated circuit chips used in the high-energy ignition systems in GM automobiles.

The chip, a 5.26-mm (0.2-in.) square, is soldered to a heat sink about four times its size. At an automatic test station, an electronic camera scans the chip. Then a computer works its way through a programmed sequence of operations with the digitized image data from the camera (see Fig. 38).

It uses edge-direction analysis to determine the approximate orientation of the chip. At the same time, it enhances corners of the chip with local template matching. Then it establishes exact location by searching for four corners that form a sequence the size of the chip at the known orientation. Finally, it verifies the chip's structural integrity (checks for broken chips, ragged edges, etc.) by detecting uniform contrast about its boundaries.

In effect, the computer calculates the position of the chip, rejects chips that are structurally defective, and directs the alignment of test probes over the chip contacts.

SIGHT-1 is the product of a cooperative effort. Dr. Michael Baird, a GMR computer scientist, developed the vision programs and demonstrated feasibility on a laboratory version of the vision system. GM Manufacturing Development converted the laboratory system into a practical production-line version meeting Delco Electronics's reliability and cost needs. This vision system was then integrated with parts-handling mech-

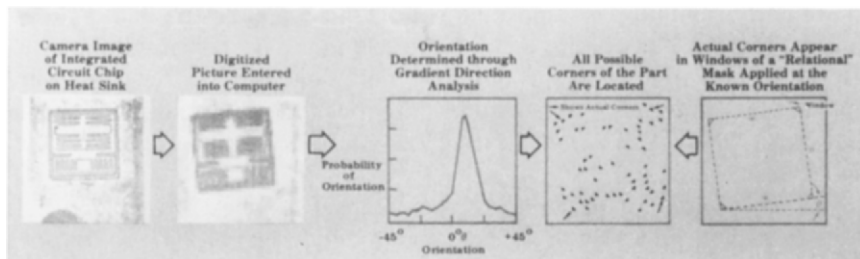


Fig. 38. Locating an IC chip.

anisms and specialized control and interface electronics designed and built by Delco Electronics to form a functional testing system for production use.

How accurate and reliable is SIGHT-1? It locates the chip with a maximum error of ± 0.105 mm (0.004 in.). The system has proved to be reliable in production environments.

c. Final Assembly. We now have group-manufactured parts, which will be assembled in a particular fashion to result in a finished product. Optimization of planning and scheduling, done much earlier by the process planner in the manufacturing engineering department, will result in a smooth flow of material to final assembly. This smooth flow of machined parts will increase the productivity at this level.

In general, final assembly is a repetitive task by nature and lends itself to the application of computer control. In this case the computer will control the motions of an articulated robot. With the integration of VLSI as the controlling unit of the robot, more intelligence will be added to the robot.

The robot first used for assembly purposes was taught its motions by a human operator leading it through its various motions. This teaching by the human operator caused the storage of coordinate points in the robot's computer memory. After the robot learns all the moves it requires for the assembly of a particular part, the computer can then read the information stored in memory and hence repeat the process over and over again.

This process works well as long as all the parts that the robot needs to pick up are oriented in the proper way since the robot has no way of detecting the parts orientation. Improper orientation of the part will cause the assembly process to grind to a halt. Now techniques are available to ensure proper part orientation, but implementation of the required hardware reduces the flexibility of the system. This loss of flexibility may not be of major concern today, especially in large batch operations, but should not be tolerated in future robot assembly systems. An example of where part orientation is important is in the placement of bolts into an assembly. If the robot were to pick up the bolt in the proper orientation, assembly would be no problem, but if the bolt picked up by the robot were 90° from the required or expected orientation, the robot would be unable to insert it. The result would be intervention by the human operator and, hence, lost productivity.

Another problem is in feedback with respect to whether the robot has actually picked up a part at a designated location or not. The robot does not know; it has been taught only where to move. Also, when the robot's "hand" grasped the part, did it possibly crush it by using too much force?

Again, both these actions are problems since the robot knows only where to go.

What needs to be developed for the robot in assembly is some way for it to see the object it is to pick up, some way to detect if it actually has picked up a part, and some way to sense the pressure its claws are applying to the part in order to pick it up and yet not crush it. Sight and tactile sense are areas where much development is needed, and VLSI along with memory enhancements can yield an intelligent, flexible robot in final assembly.

An example of such a robot is the PUMA (programmable universal machine for assembly) robot used by General Motors (see Fig. 39). In the laboratory demonstration depicted, the PUMA is installing light bulbs in the backs of instrument panel cluster cases. The PUMA robot is seen as part of an assembly system that would include parts feeders, conveyors, and people. General Motors's first production installation of a PUMA robot was late in the spring of 1979 at the Delco Products Division, Rochester, New York.

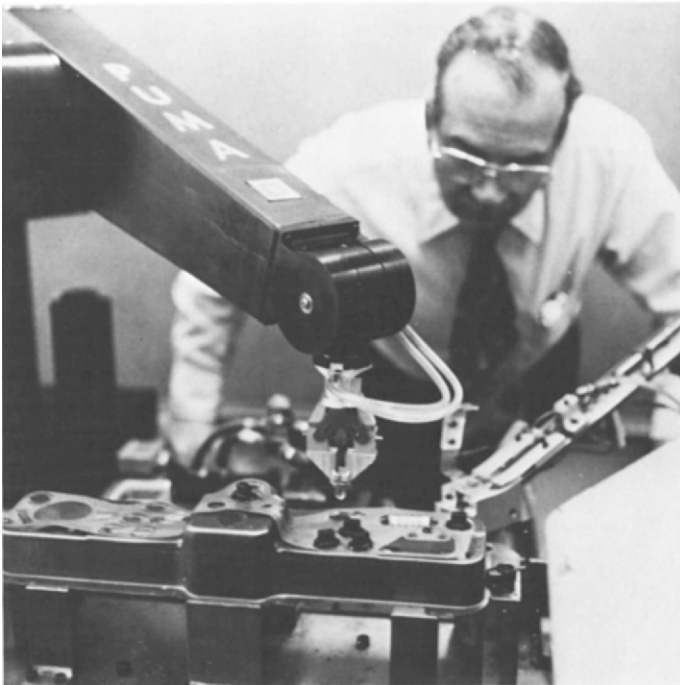


Fig. 39. PUMA in operation.

Sight in the form of image processing will alleviate the problem of part orientation. The part can be presented to the robot via a conveyor system and, irrespective of its orientation, the robot's vision system will allow the robot to pick up the part in the correct orientation. A vision system will also allow the robot to discriminate among a variety of parts. Thus component parts for the assembly can arrive on the same conveyor system and the robot's vision system can not only discern part orientation but also discriminate among different parts and choose the parts in the proper order needed for assembly.

Another system developed at General Motors is the SIGHT-I system. A system like SIGHT-I (see Section III.D.2) does a good job if you want to locate only one type of part, but automotive part transfer operations require that systems be able to locate many parts and be easily taught to recognize new parts. For that a more sophisticated system is needed.

Enter SIGHT-II, a computer vision system that GMR is now developing to meet these requirements. Show this system a new part, a connecting rod, say, under good lighting conditions and it will analyze the part, compute its geometric properties, form a model, and commit the model to memory. It then finds similar connecting rods by matching their characteristics with those of the stored model (see Figs. 40 and 41).

Specially developed programming allows it to find a rod amidst a collection of different parts, even if the rod is poorly lit, only partly visible, or has a low-contrast background.

For automatic loading of machines, it is often necessary to know which side is up. In Fig. 41, an experimental computer vision system makes this determination for a connecting rod. The main point of the distinction between the two sides of the rod is the camlike lobe located around the wrist pin center on one side. The lobe is too small to be reliably detected by mechanical sensing probes. In this case, the rod must be in a fixed position in the viewing area. "That simplifies things," explains Steven Hol-

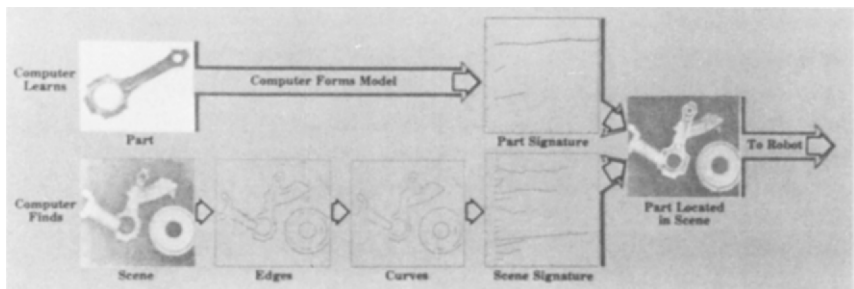


Fig. 40. The computer learns to find a connecting rod.

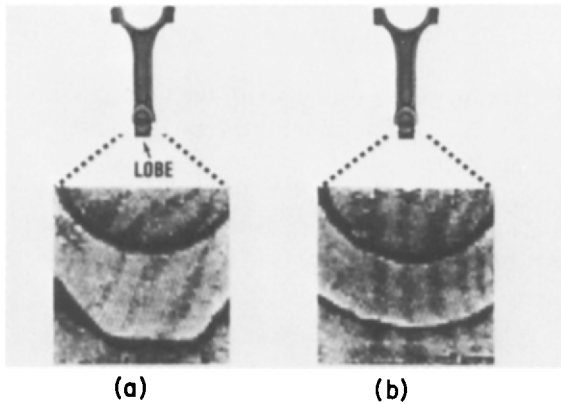


Fig. 41. The computer learns which side is up (see text for discussion): (a) with lobe; (b) without lobe.

land, the computer scientist who worked on the problem. “We adjust the camera’s field of view to include only the small area of the lobe.” Then, by measuring the width of the lobe, the computer knows which side is up.

Dr. Walton Perkins, a senior research physicist, developed the computer programs for this trainable system. He points out, “Here the model is separate from the recognition component and is not built into the program. So the model can be readily changed when you need to locate a different part. You don’t have to reprogram it.”

Currently, GMR is working with GM Manufacturing Development and Central Foundry Division to adapt SIGHT-II to production applications. It now runs on a large computer and takes 15–20 sec to find a part, which is too slow for many production-line applications. The immediate goal is to convert it to run on minicomputers and cut its operation time down to 3 sec or less.

Some hard problems remain, and as project leader Lothar Rossol comments, “Before computer vision revolutionizes manufacturing processes, we have at least two technological obstacles to overcome: the expense of computing—picture processing is often a costly proposition—and our still feeble understanding of the mechanisms of perception. But we believe these obstacles are being surmounted.”

The group is involved in research in many other areas in addition to those mentioned here, areas such as automatic programmable assembly of parts, methods of picking jumbled parts out of bins, improvements in the control of robots, and computer sensing of color.

“Not so many years ago the idea of providing machines with perceptual

capabilities was considered science fiction," notes Rossol. "With the building of SIGHT-I, SIGHT-II, and other systems, yesterday's fiction is becoming today's fact."

Robots in production plants today are of the first-generation type. They cannot see or feel. They move along a preprogrammed path that never changes, even if the part is out of position. The next generation of robots will be able to see or feel, and they will react to what they see or feel. Their path will change on each cycle, as the position of parts change or as the requirements of the job change. Current research by General Motors



Fig. 42. Placing a part on a conveyor belt.

Laboratories's scientists is aimed at developing computer vision systems that will allow robots to locate parts that are overlapping or are partially obscured and in mixtures of different kinds of parts. In the laboratory demonstration shown in Fig. 42, Lothar Rossol of the GMR Computer Science Department is shown placing a part on a conveyor belt. A television camera mounted in the apparatus at the rear views the moving conveyor belt below it and the computer forms a digitized picture of the part (a representative picture is shown in the inset). The computer goes through a series of recognition techniques and, when it makes a match with a similar part already in its program, advises the robot arm where the part will be on the belt so that it may pick it up.

Picking up moving parts from a conveyor belt and stacking them until they are needed normally is a simple assembly-line task, but an industrial robot that is virtually "blind" would find the job overwhelming. Scientists at the General Motors Research Laboratories trying to provide "sight" for industrial robots report they have developed optical systems to aid

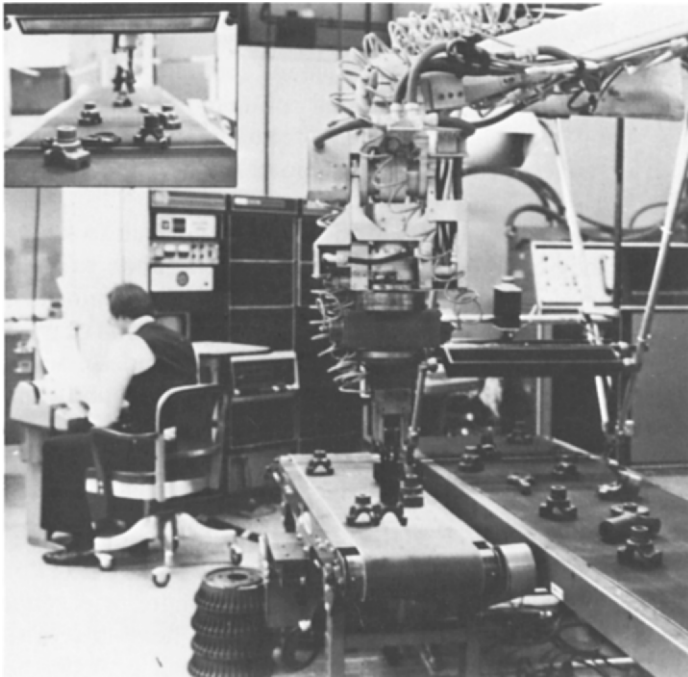


Fig. 43. Demonstration of robot selection of part.

computers in locating parts on the conveyor belt. The demands placed on the computer increase significantly as the scenes to be processed become larger and more complex. The use of optical processing to filter out all but the essential information needed by the computer—the location of the part on a moving belt—is one of the methods being explored to aid the computer vision process. In this laboratory demonstration (bottom of Fig. 43), a white light illuminates the area on the conveyor belt. As the part, in this case, a connecting rod, moves through the light, its image is fed through an optical processor (center background) and then through a computer that “tells” the robot the part is on the belt and what action to take to pick it up. Dr. Aaron Gara of the GMR Physics Department (Fig. 44) is shown with the optical processor that forms a laser image of the scene, compares the information in this image to a prerecorded optical signature of the part, and, when correlation occurs, produces the signal that provides the robot arm with the coordinates of the part on the conveyor belt.

This large industrial robot (Fig. 44), equipped with vision capability, is undergoing further laboratory development and testing at GM Manufacturing Development, GM Technical Center, in Warren, Michigan.

Four major elements of the system are:

- (a) the robot unit and its control system, including specially modified software to permit acceptance of visual commands;
- (b) the vision element consisting of the lighting system (“line of light”; inset at top left), camera, and computer, where the image is stored until used;
- (c) a motion sensor or encoder on the conveyor belt to keep track of the parts after they have been seen by the camera; and
- (d) the supervisory computer, which takes all of the information provided by the camera (shape, size, location, and orientation of parts traveling on the belt), the belt encoder, and the robot, and so on, and then directs the robot element to pick up the parts and properly dispose of them, according to a preprogrammed scheme.

A system such as this would lend itself to material handling, transferring and packaging, or palletizing of larger, heavier parts.

A lot of research needs to be done in the area of sight and tactile sensing for the robot before its full potential is realized. Work is being done, as is exemplified by the previous examples, but much more research needs to be done in order to increase the productivity and flexibility of robots in final assembly. Both VLSI and denser memory will provide the power needed to control the robot, the sight, tactile sense and communicate between itself, and a larger monitoring and controlling system integrating the work of many robots.



Fig. 44. Optical processor (top) and industrial robot (bottom).

IV. EDUCATION AND TRAINING

A crisis of attitude has existed for several years with respect to manufacturing, and it has resulted in much talented manpower being shifted away from manufacturing toward the more glamorous industries of space, computers, and microminiaturization. This loss has been not only with respect to engineering but also in skilled craftsmen such as machinists. It is important to emphasize that the manufacturing technologies discussed herein incorporate some of the generic and underlying technologies that

are so much a part of the more glamorous industries. These include VLSI, of course, robotics, automation, and the use and implementation of microprocessors. As such, there is a general requirement for the education of the public as well as the young engineers coming out of the universities and colleges of this country to understand and appreciate the modern concepts and approaches involved in the manufacturing technology.

Universities and colleges need to introduce young engineers to manufacturing. During this training, they should be shown by hands-on experience the challenging and interesting careers that are available in these related fields. There is also an increasing obligation on the part of the universities and colleges for reeducation or retraining. This need will continue to increase as the rapidity of changing technology increases.

This problem of education and training and the sense of awareness of manufacturing and its importance to the health of a country becomes the joint responsibility of government, private industry, and the academic sector. As examples of the recognition of this interdisciplinary role, the United States government has already established five national centers under government sponsorship, which includes one in the area of robotics. Another example is the government sponsorship of the submicron facility at Cornell University supported by the National Science Foundation.

Industry, also recognizing its role in this complex educational and training process, has taken its own initiative as well. This is exemplified by the integrated systems operation at Stanford University, which has gained strong industrial support, as well as the VLSI facility at the University of Minnesota, again sponsored heavily by industry. A third example is the industrial support by the General Electric Company, Boeing, and General Motors of the Center for Manufacturing Productivity at Rensselaer Polytechnic Institute.

With slightly more detail, this last example can be used to demonstrate the operation of an industrially sponsored center in an academic environment. It is operated and run by a full-time administrative and technical staff occupying nonfaculty positions. Projects are performed for industry on a self-liquidating financial basis by the center's full-time staff using graduate and undergraduate students in concert with university faculty to accomplish the end goal. These projects enable students to gain experience they otherwise would possibly never have had.

Regarding reeducation, it is anticipated that this will become a major aspect of the educational responsibility in the future. The people in the shop, from machine operators, maintenance personnel, engineering, manufacturing engineering, and management-level supervision, need to be reeducated on a fairly regular basis. They all need to learn and appreciate

the new devices and systems in order to use them properly to their fullest potential and maximize the productivity gains that can be realized. It is expected that this training will be a combination of formal as well as informal programs run at established universities and colleges as well as in-plant courses and seminars. This will further promote the synergism between industry and the university, which is so vitally a part of the success of this field.

The large expense incurred with VLSI equipment and facilities should not go unmentioned. In the area of processing, the trend has obviously been from hot and wet processing to cold and dry processing. A computer-aided design approach now realizes the integrated circuit by direct write capability on the wafer using a sophisticated submicron e-beam facility. Such equipment, facilities, and capabilities are expensive and have relatively short technical life because of the rapidly changing technology. As such, industry and the university must cooperate in the sharing of resources and laboratory facilities so both may benefit for the common good.

There is probably no field that will be more revolutionized by the impact of VLSI than computer-aided manufacturing. One looks to the implementation of VLSI in the areas discussed within this chapter and the implicit impact of VLSI in the computer area and the ramifications of the computer in the CAM field. Figure 45 graphically shows manufacturing-related jobs that have been impacted by the introduction of computers and require either new skills or the creation of new positions. The change

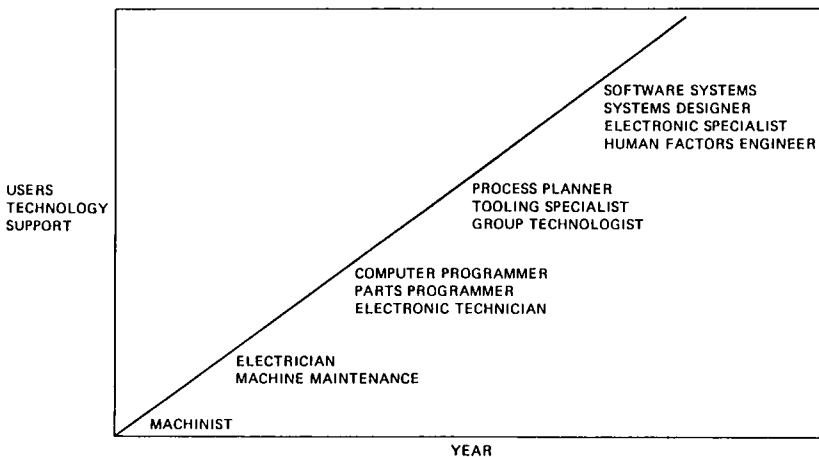


Fig. 45. Growing support requirements (courtesy of Dr. R. Miskell, Machine Tool Task Force Conference).

that will be brought about by the impact of VLSI on computer-aided manufacturing will require a new breed of engineers with backgrounds in many disciplines. They will not be restricted simply to mechanical engineering but will heavily involve electrical and computer engineering as well as materials and chemical engineering.

Investment in the human element in terms of basic and advanced education as well as continual upgrading by retraining will develop the knowledgeable people who will readily accept new technology rather than oppose it, who will implement it rather than let it sit idle, and who will hasten innovation in the computer-aided manufacturing field rather than impede it.

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